

Section 1:

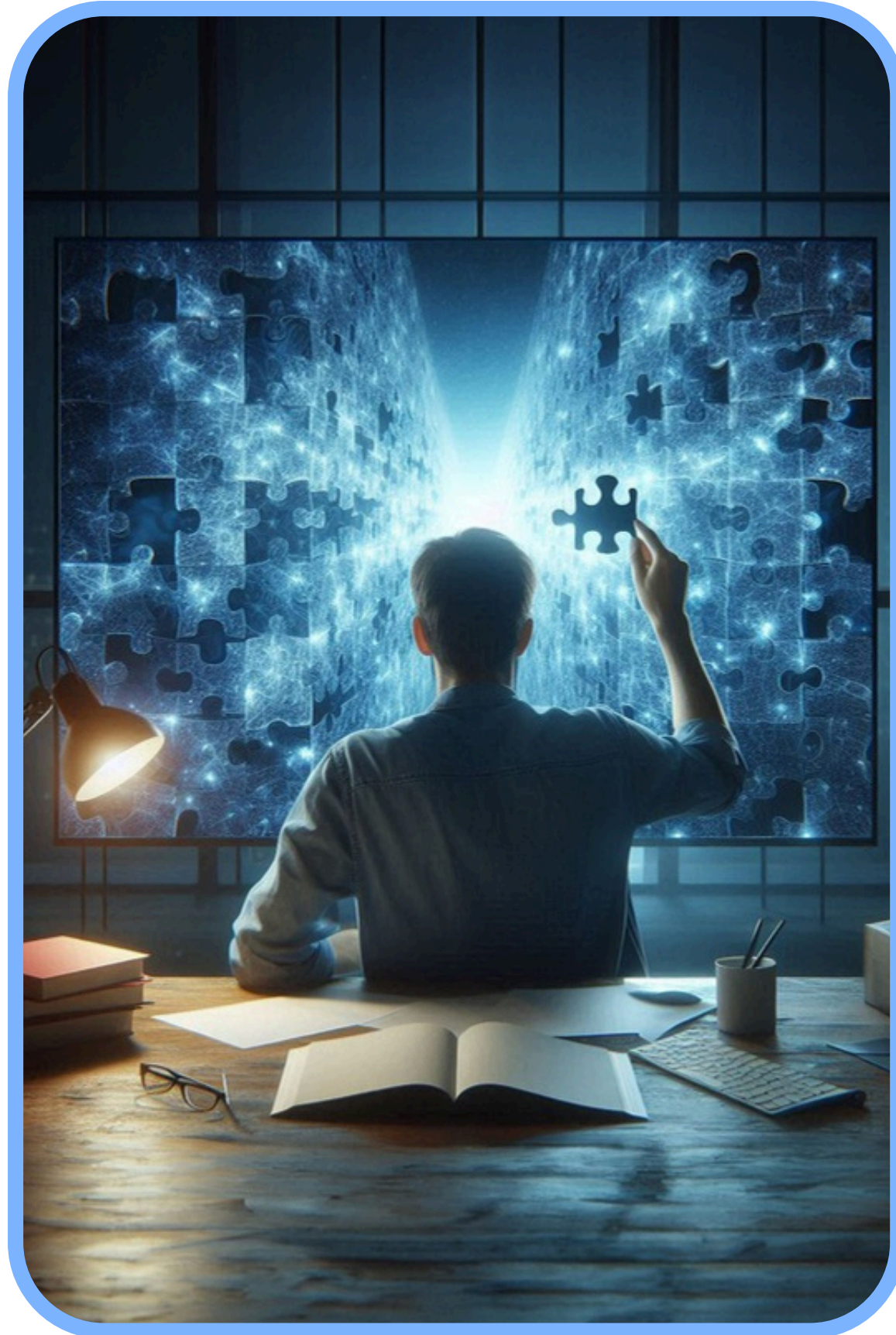
Introduction

Roles in the Software Development Team



Requirements Engineer

A Requirements Engineer is a professional responsible for **eliciting, analyzing, documenting, validating, and managing** software/system **requirements**. Their role ensures that stakeholder needs are understood, agreed upon, and translated into clear, actionable specifications that guide system development and testing.



Requirements Engineer

- A Requirements Engineer is mostly a Product Owner or a Business Analyst.
- They act as the link between the client and the IT team, ensuring that business needs are translated into technical requirements.

UI & UX Designer



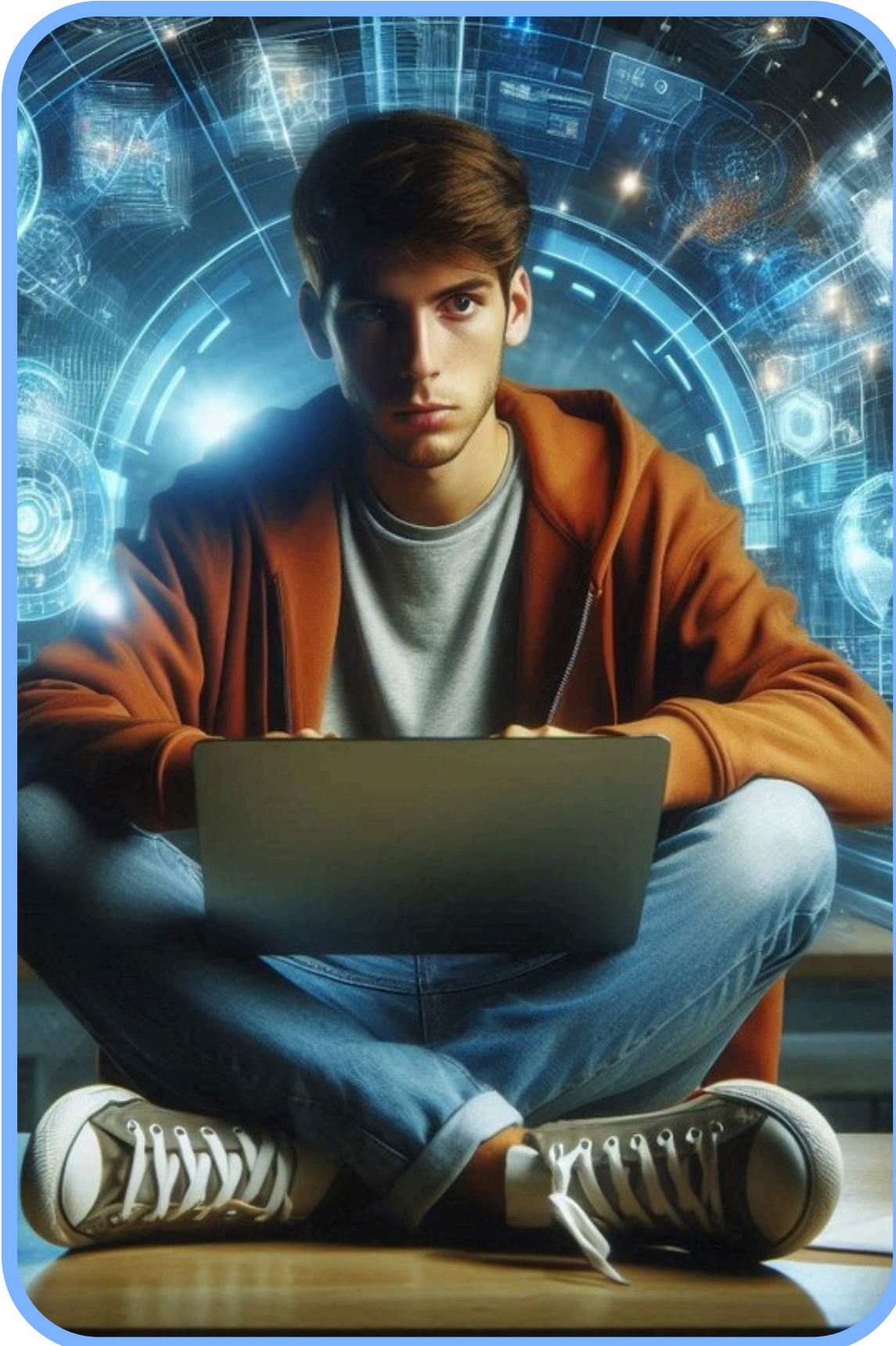
A UI/UX Designer is a professional responsible for creating user-friendly, visually appealing, and efficient digital experiences.

- **UX (User Experience)** Design focuses on researching user needs, defining user journeys, and ensuring usability to make products intuitive and valuable.
- **UI (User Interface)** Design focuses on visual elements, layout, typography, and interaction design to make products attractive and easy to navigate.

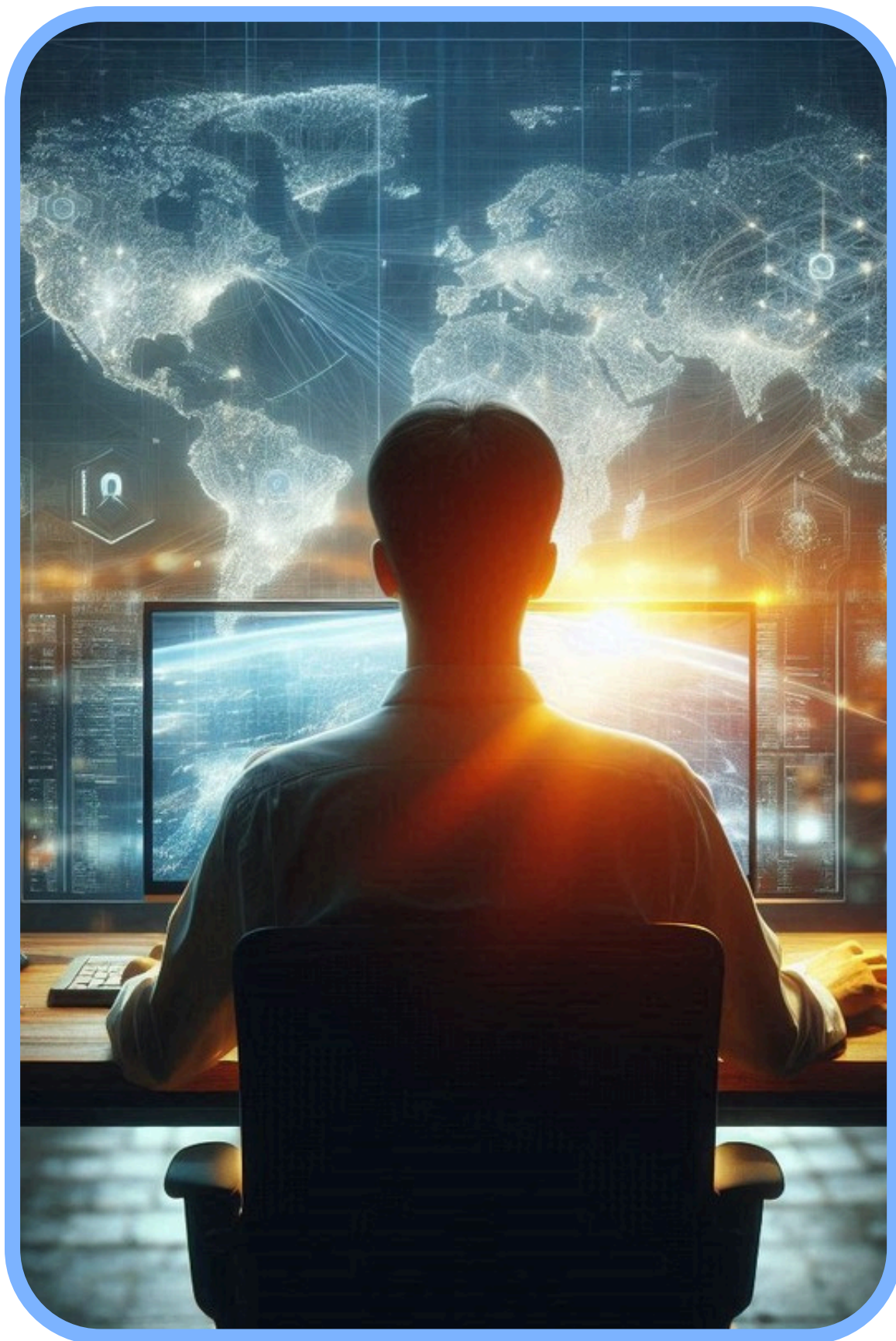
👉 Together, a UI/UX Designer ensures that digital products are both functional and delightful to use.



Front-End, Back-End, & Mobile Developers

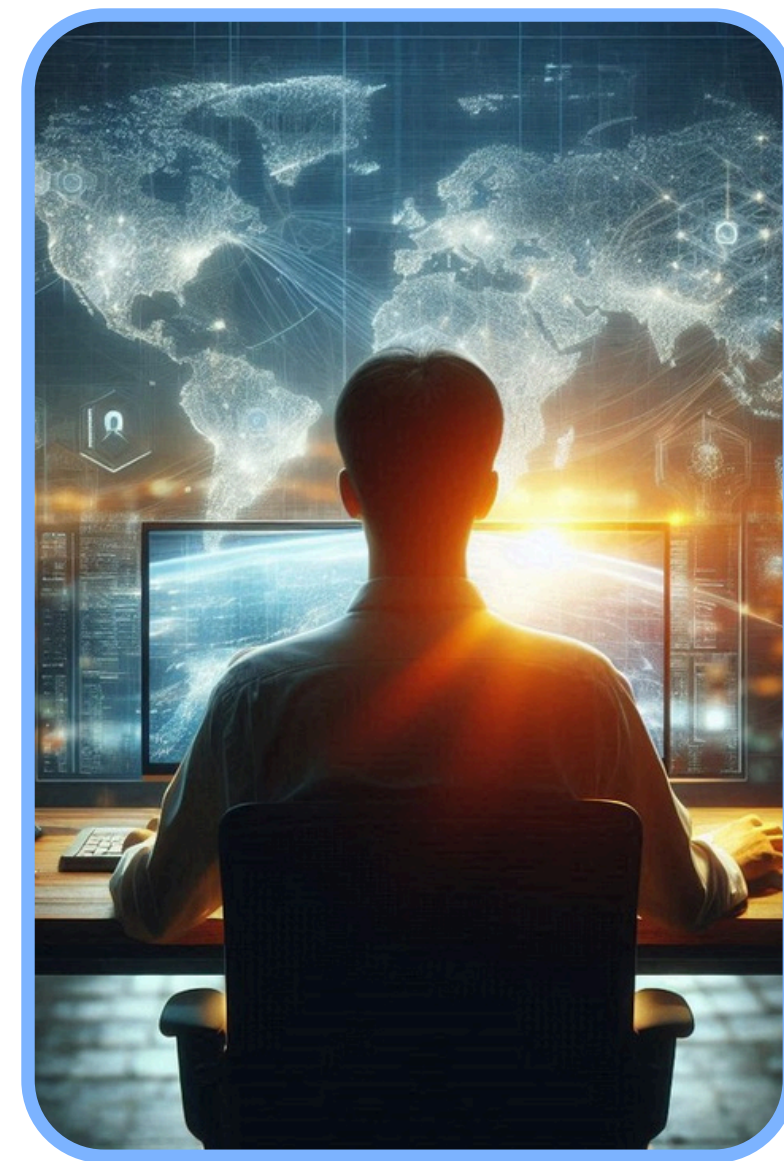
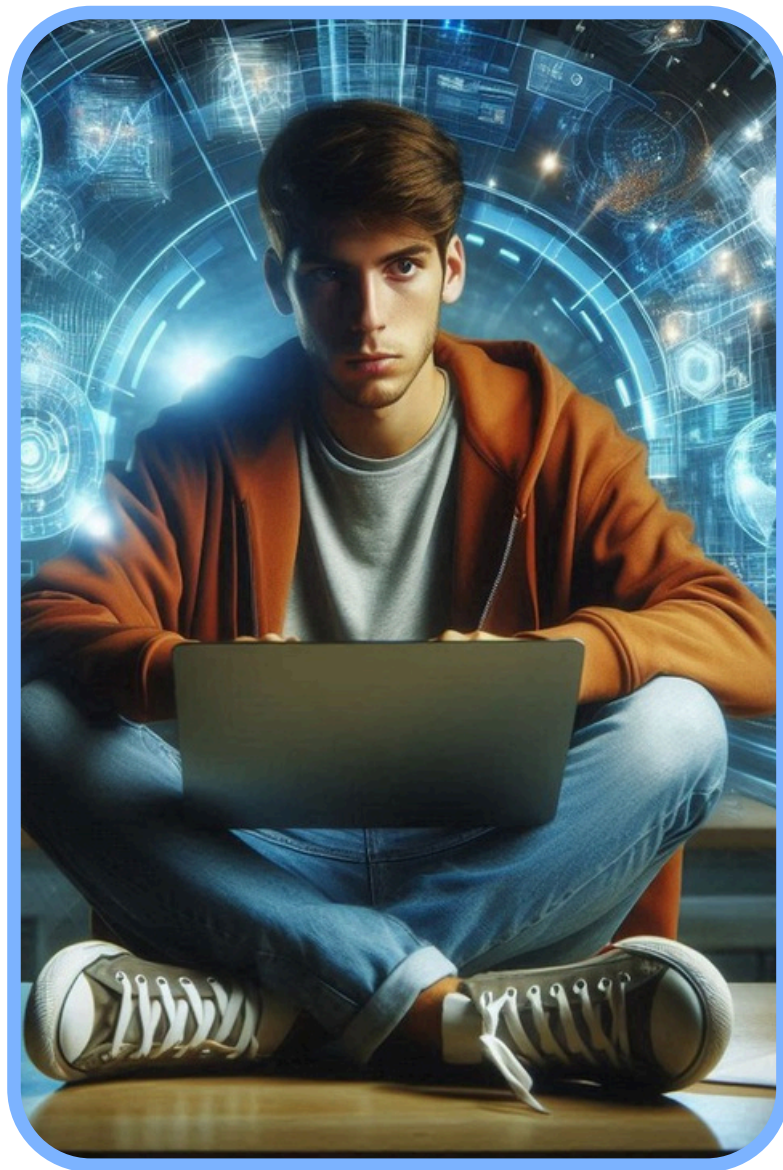


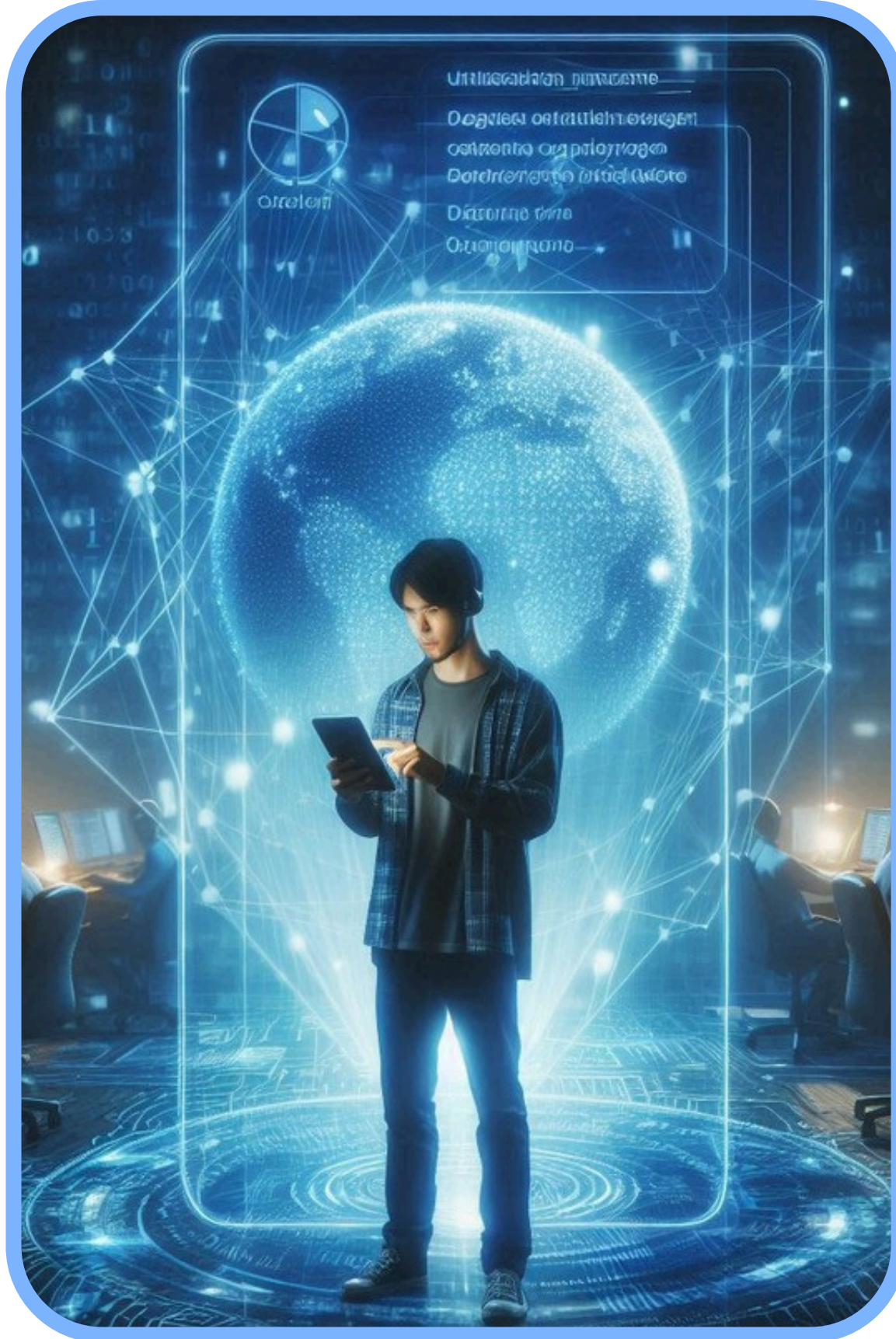
Front-End Developer



Back-End Developer

Full-Stack Developer





Mobile Developer



Project Manager

It's not that simple

- Data scientist
- Database Administrator
- AI Specialist
- DevOps Engineer
- Ethical Hacker
- Cloud Engineer

How Software is Developed

Software Development Lifecycle Models

Sequential Development



Analysis



Design

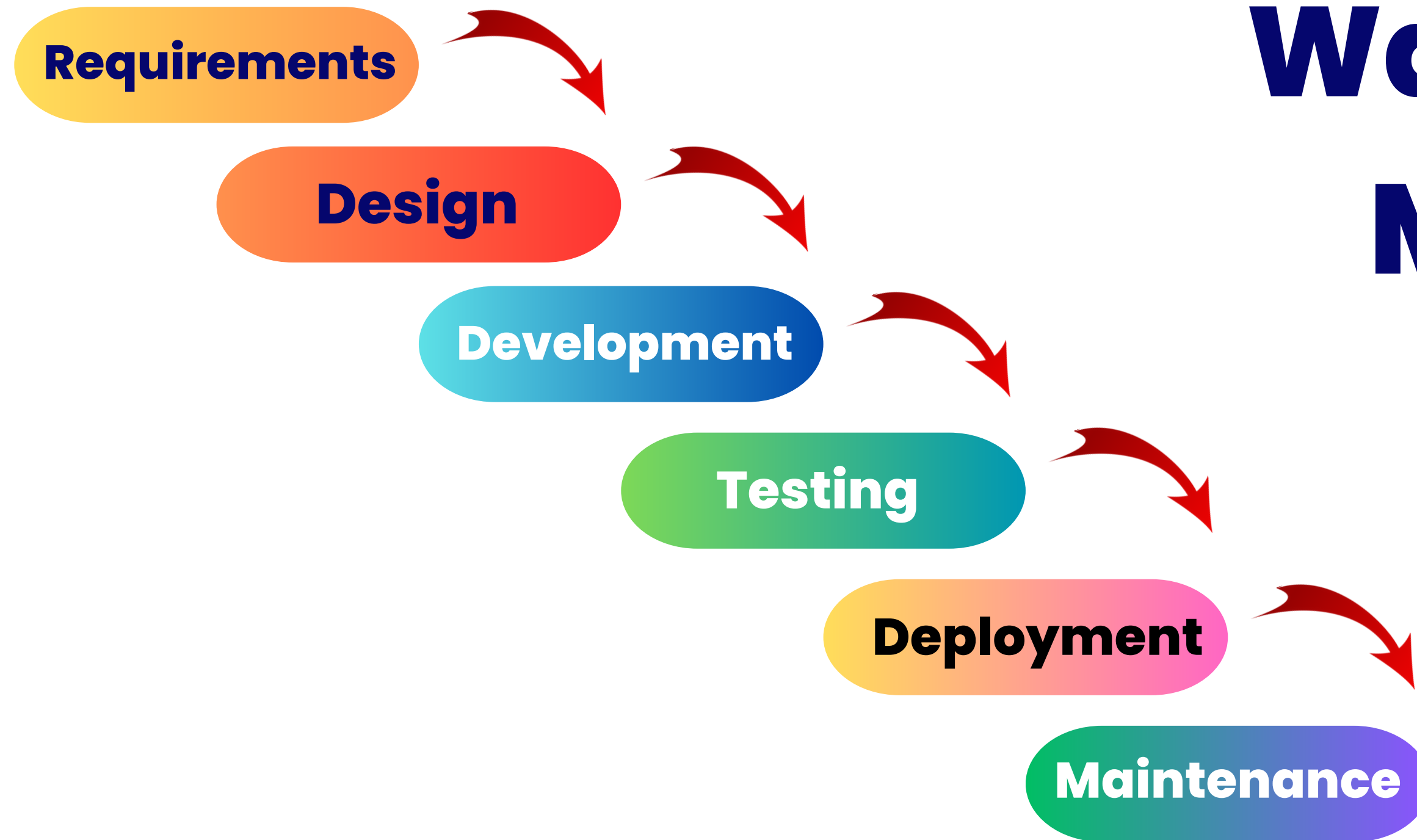


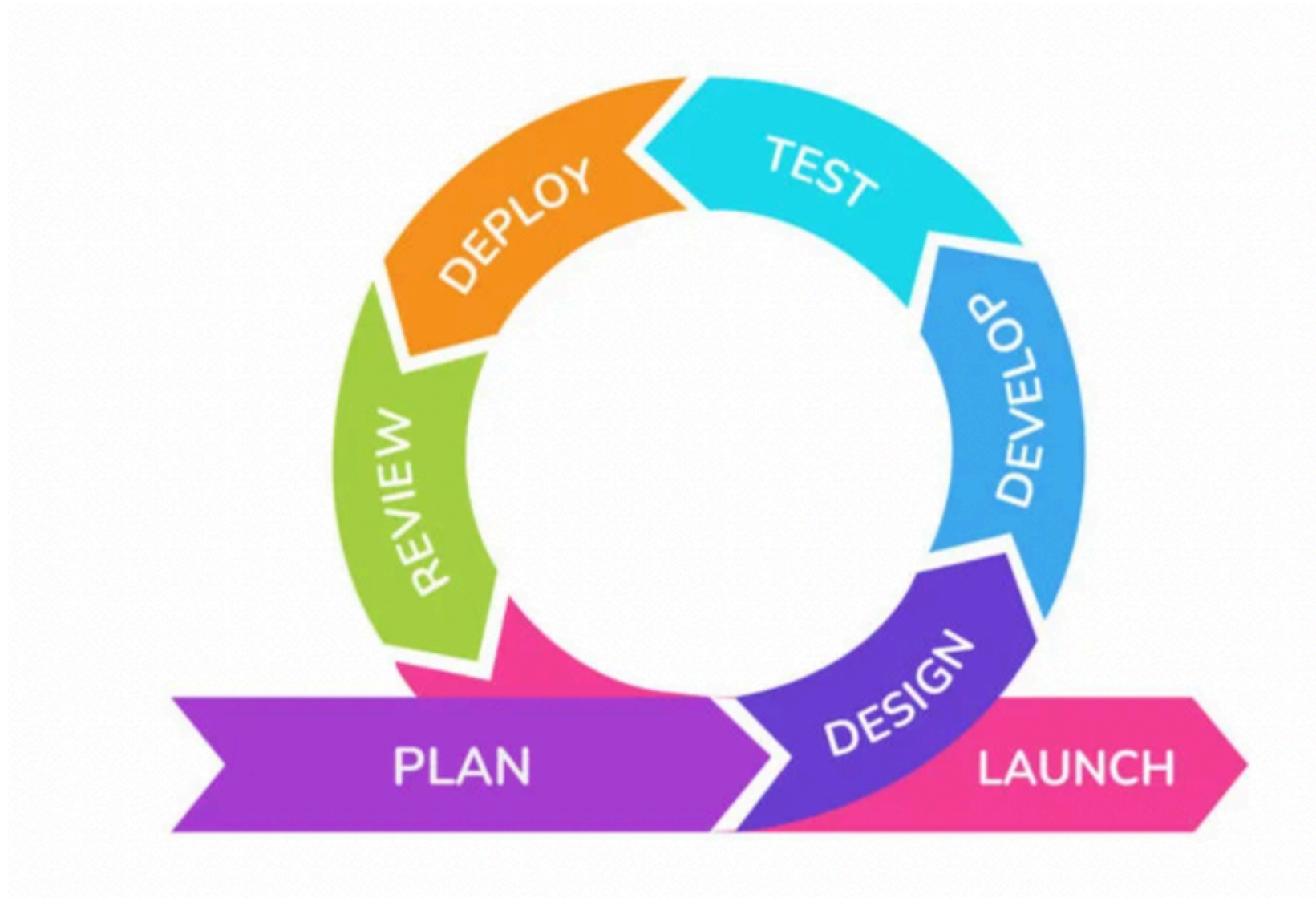
Implementation



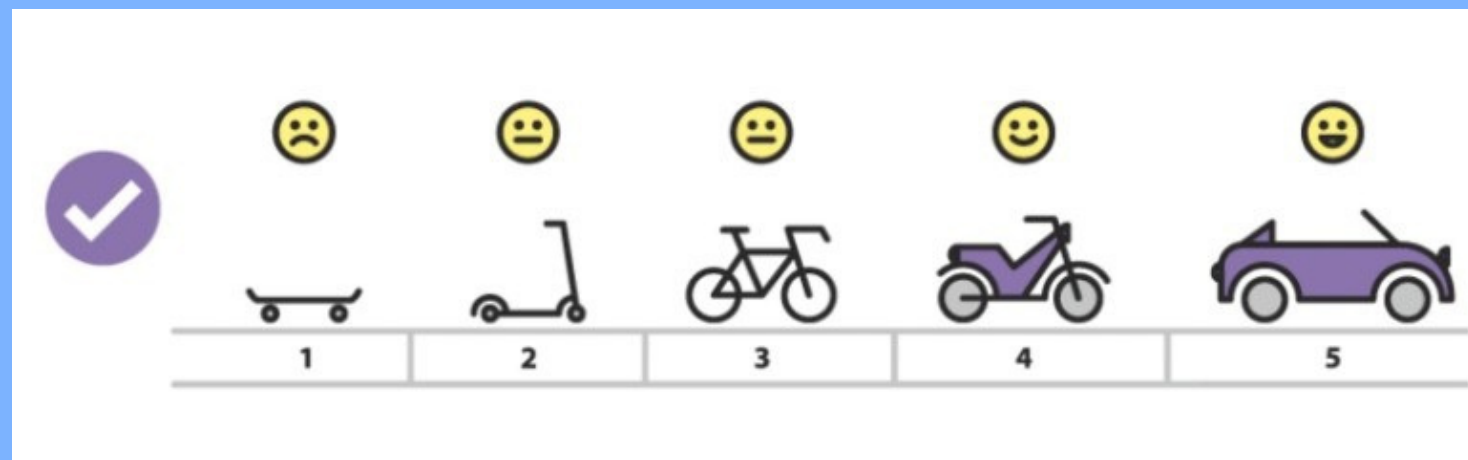
Testing

Waterfall Model

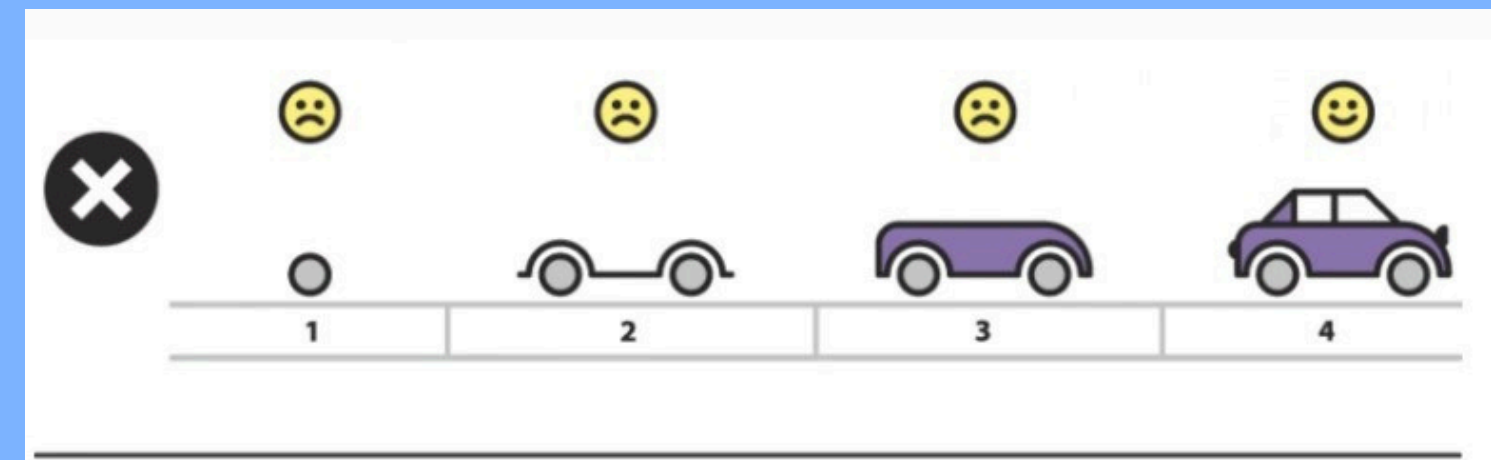




Agile Software Development

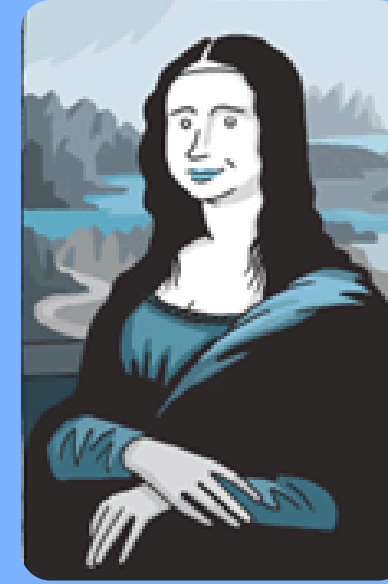
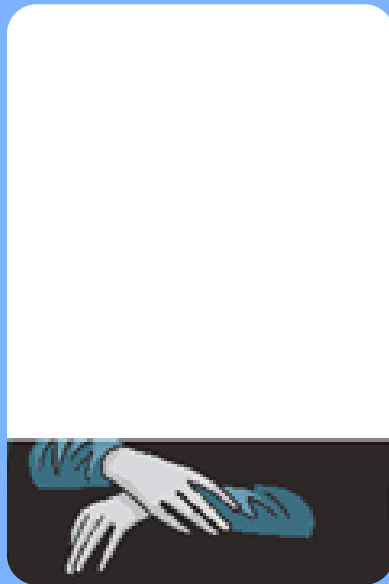


Iterative

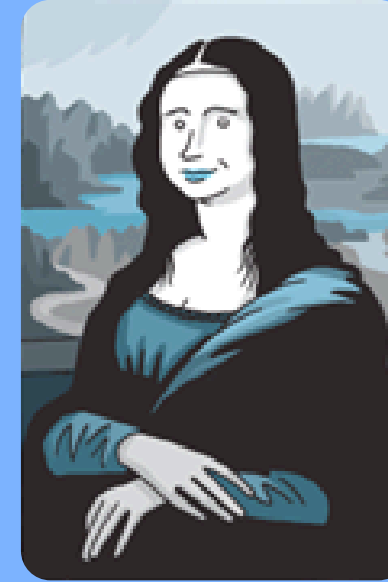
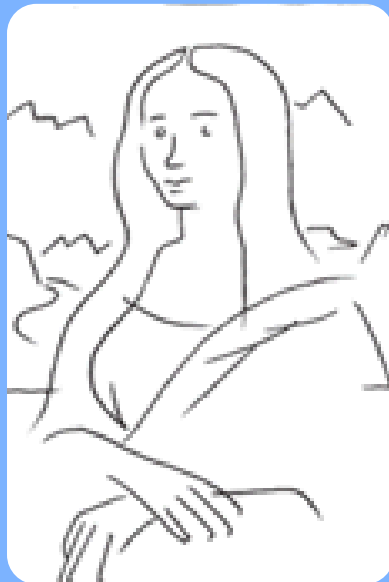


Incremental

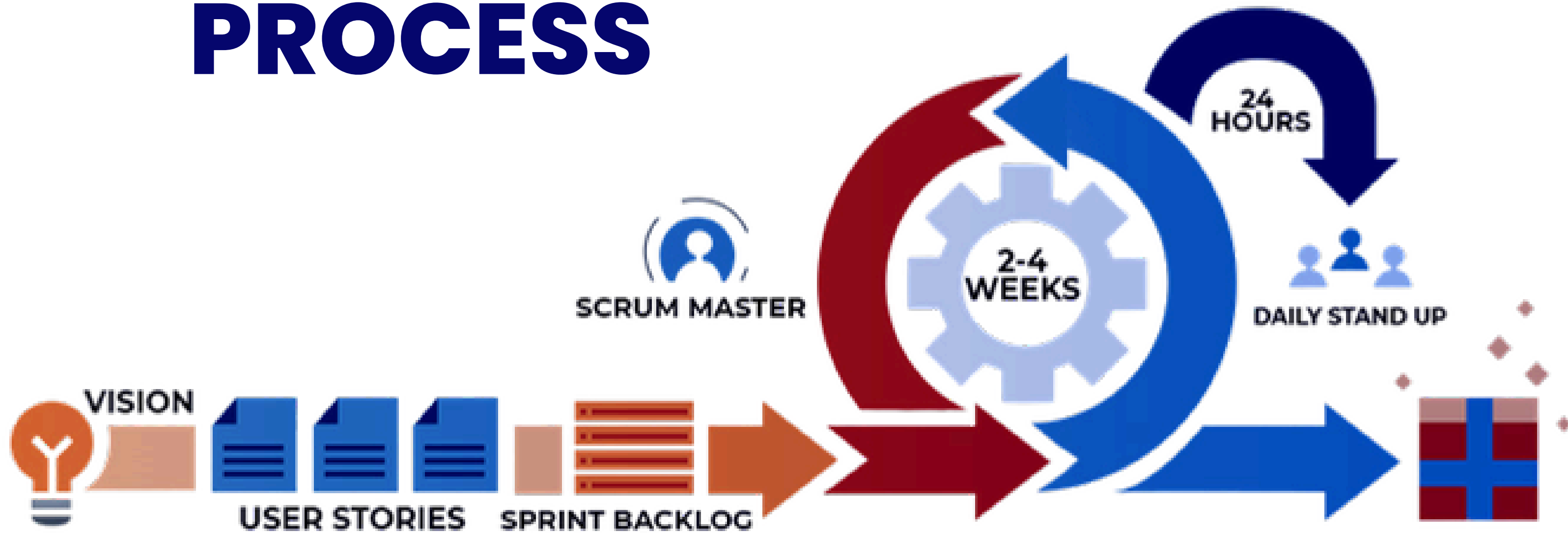
Incremental



Iterative



SCRUM PROCESS



Basic Concepts of Software testing



What is Software Testing

What is Software Testing ?

- Most people have had an experience with software that did not work as expected.
- Software that does not work correctly can lead to many problems, including:
 - loss of money, time, or business reputation
 - Injury or death

What is Testing ?

Software testing is a way to:



assess the quality of the software

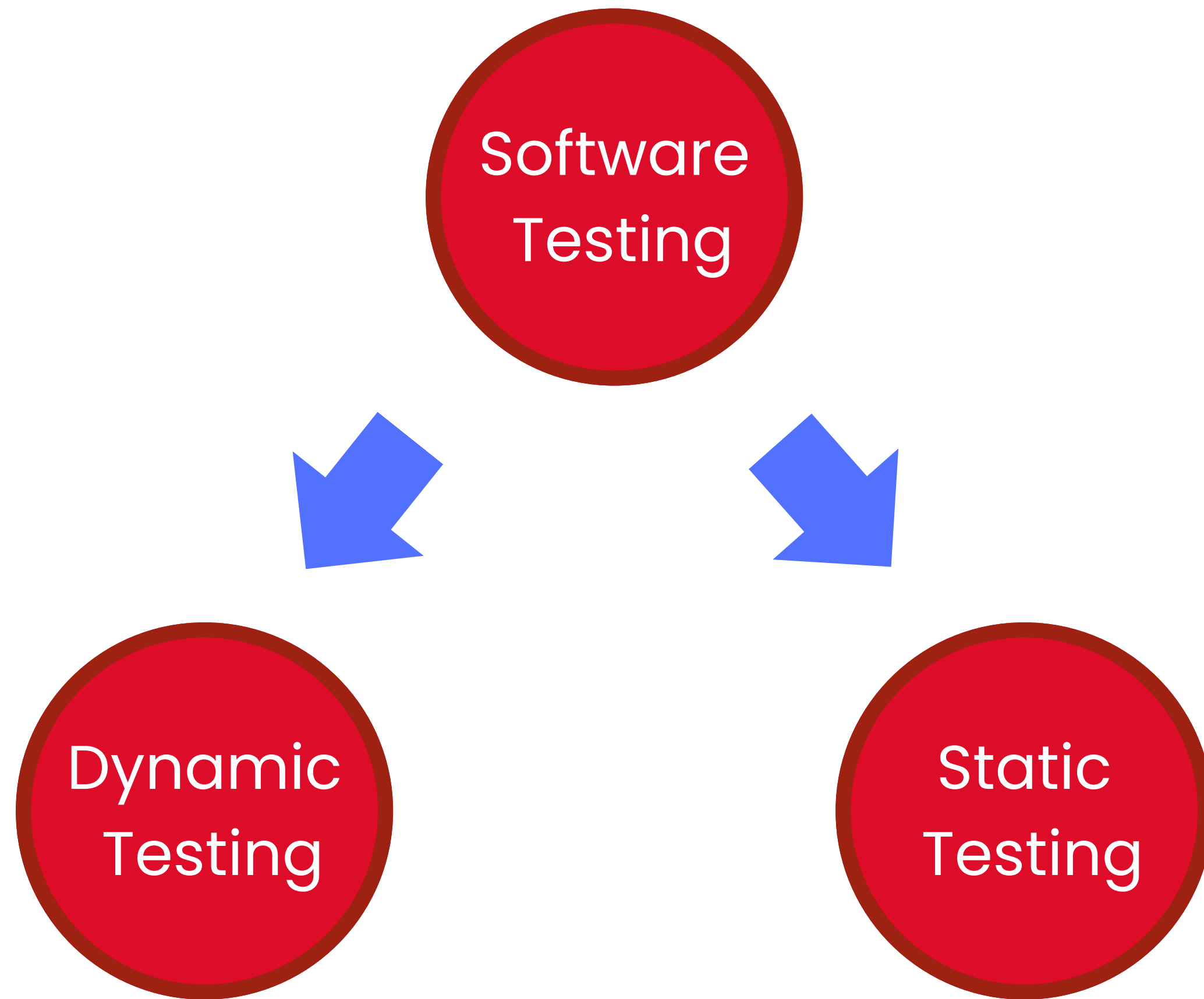


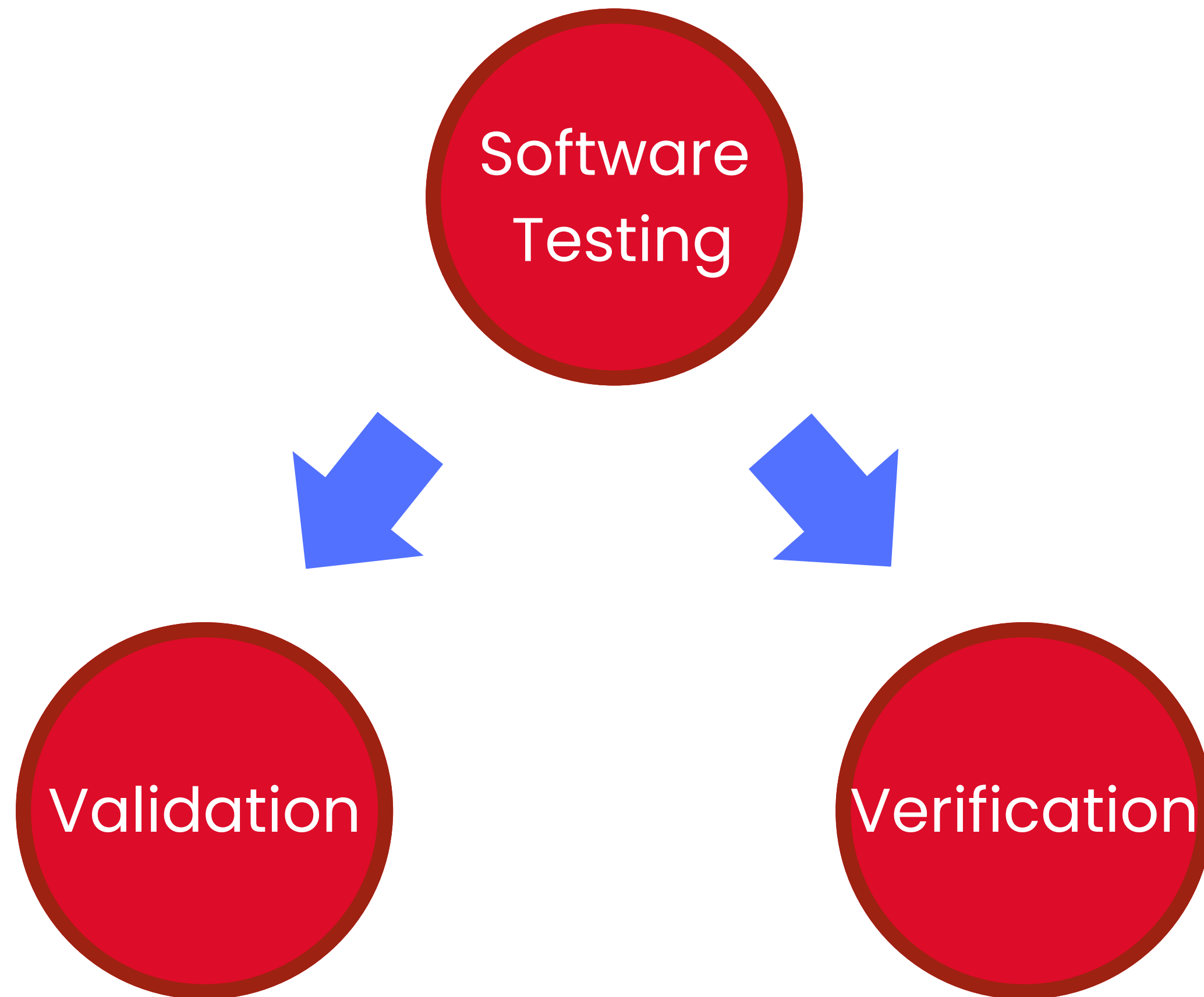
reduce the risk of software failure

in operation

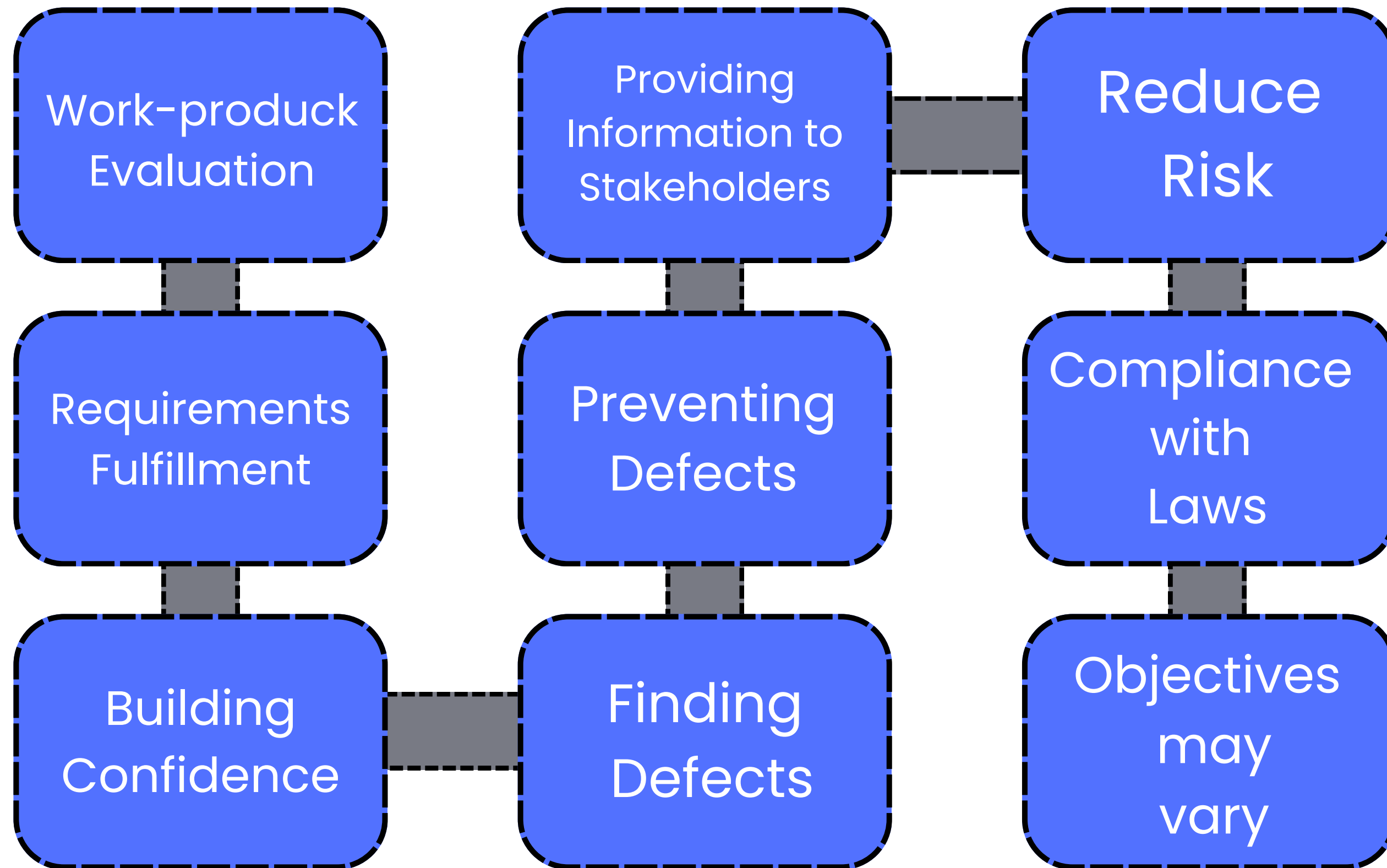
What is Testing ?

- Software Testing is not only Test Execution
- software testing **is a process** which includes many different activities
- Execution is only one of these activities.





Objectives of Testing



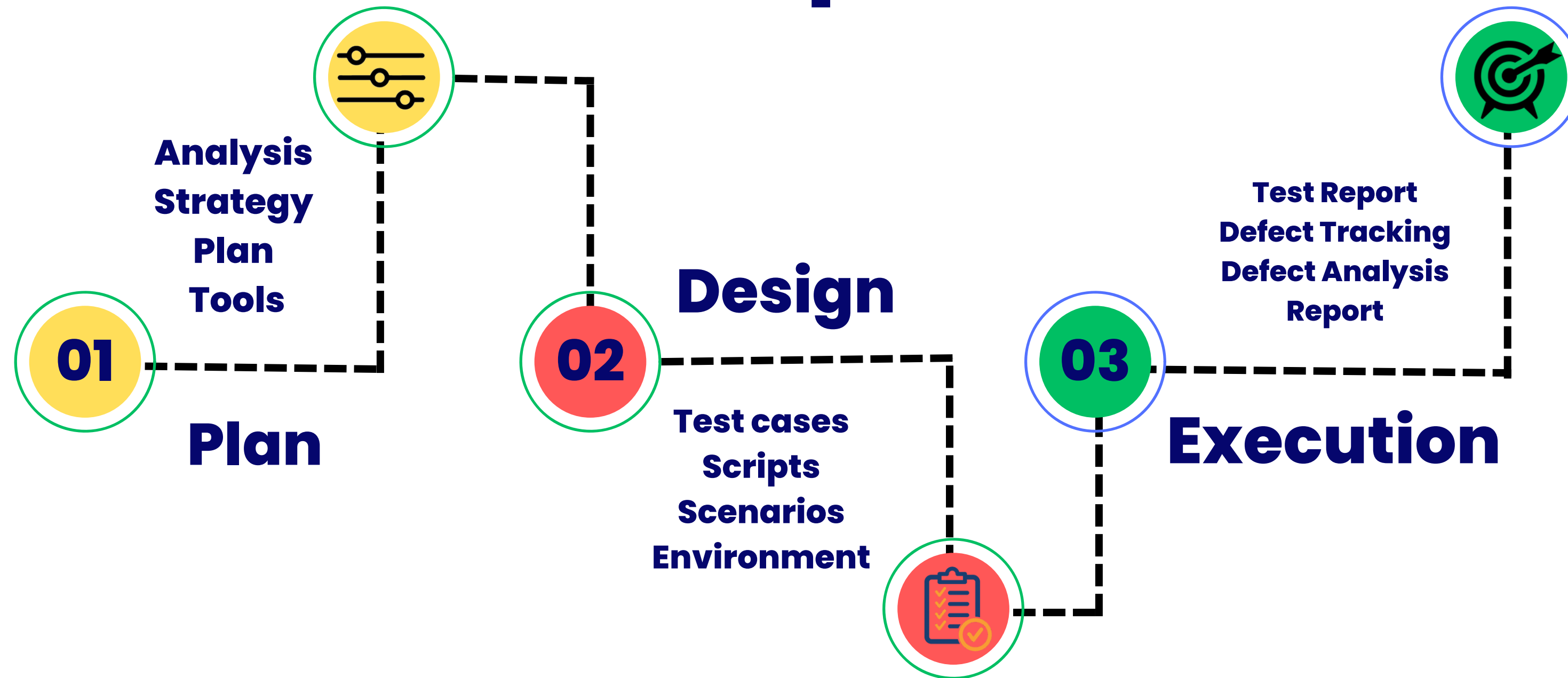
Testing & Debugging





Test Process

Test process



There is no one universal software test process, but there are common sets of test activities without which testing will be less likely to achieve its established objectives

Test process Activities

**Test
Planning**

**Test
Monitoring
& Control**

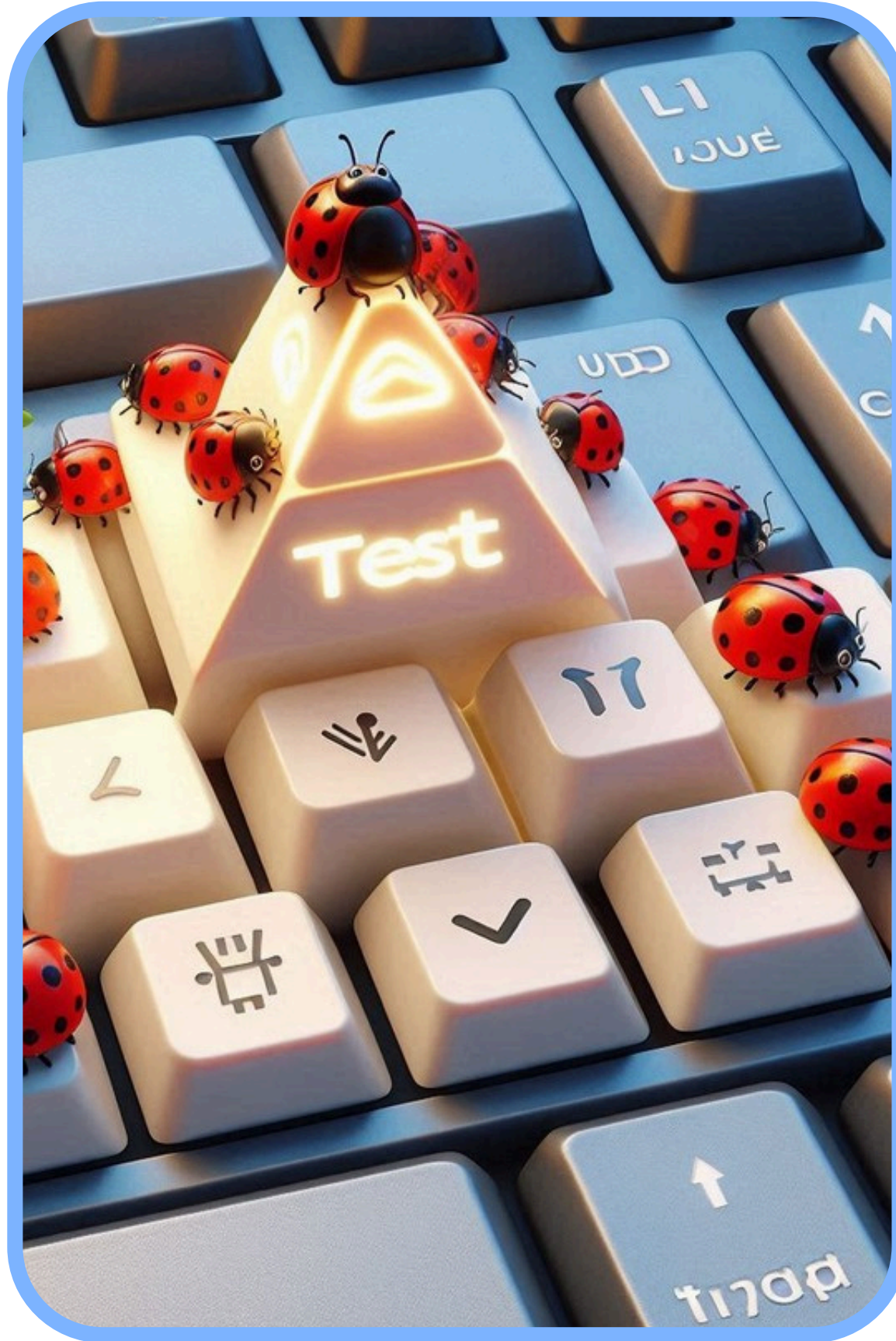
**Test
Analysis**

**Test
Design**

**Test
Implementation**

**Test
Execution**

**Test
Completion**



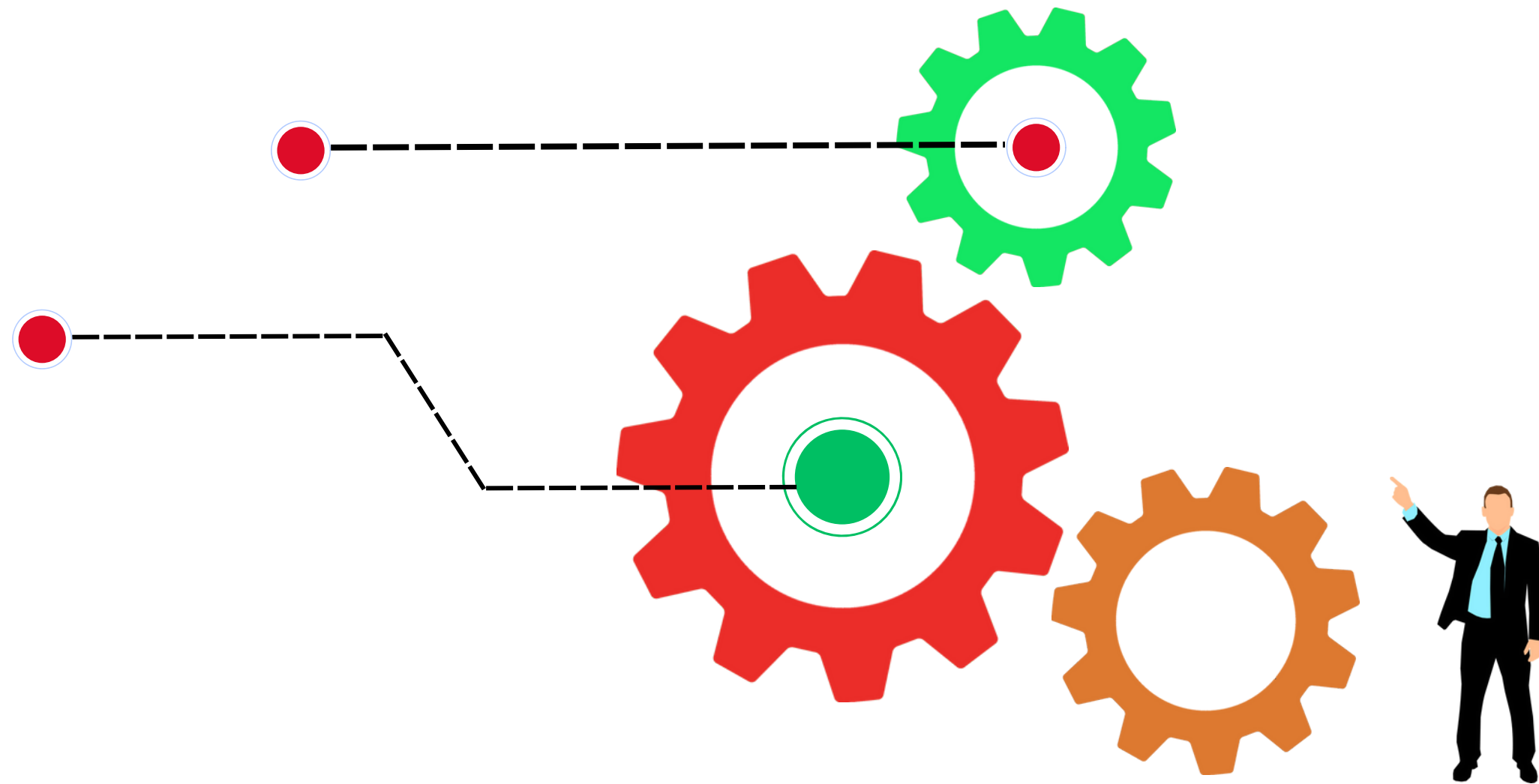
Test Levels

Component Testing

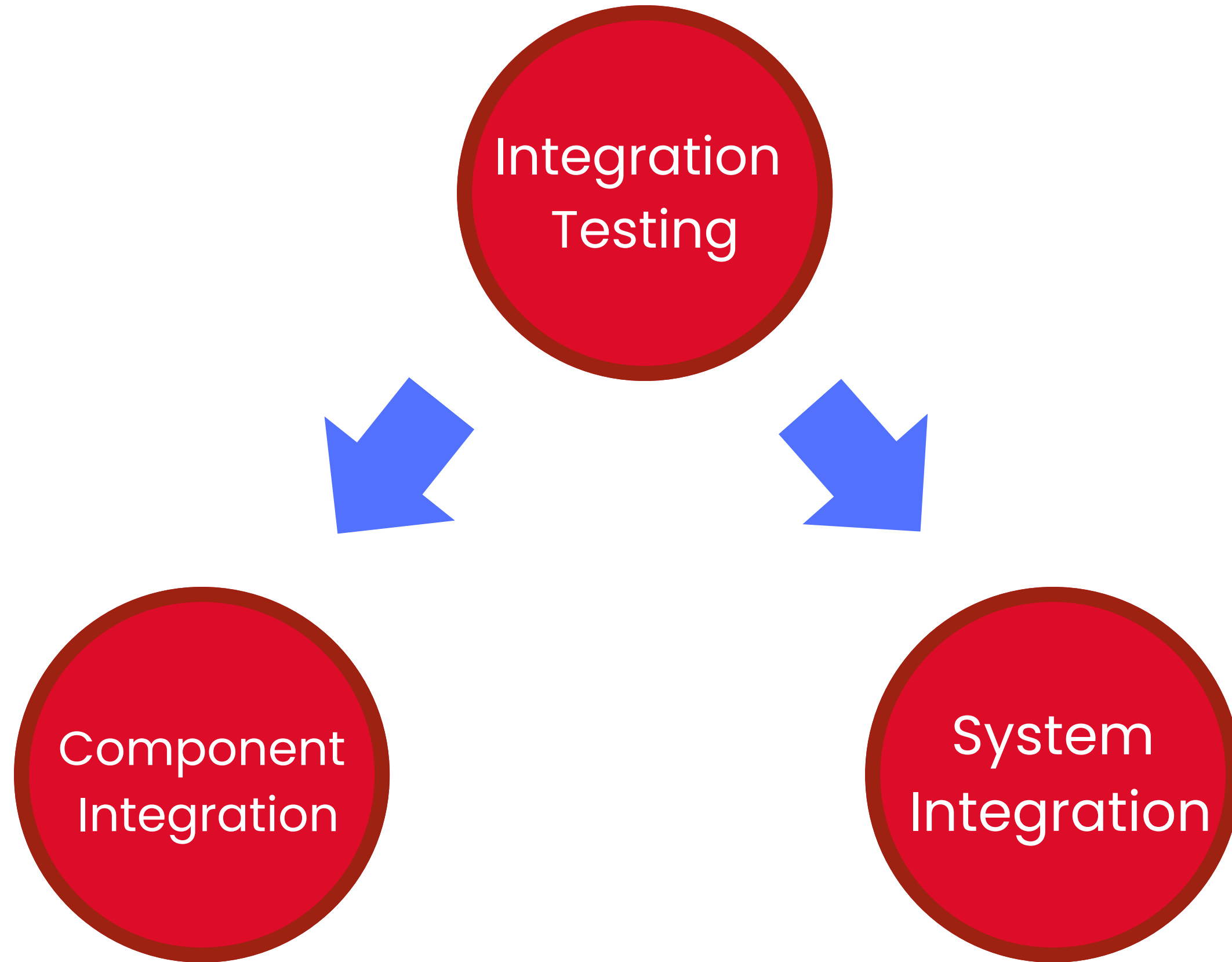


Component testing (also known as unit or module testing) focuses on components that are separately testable

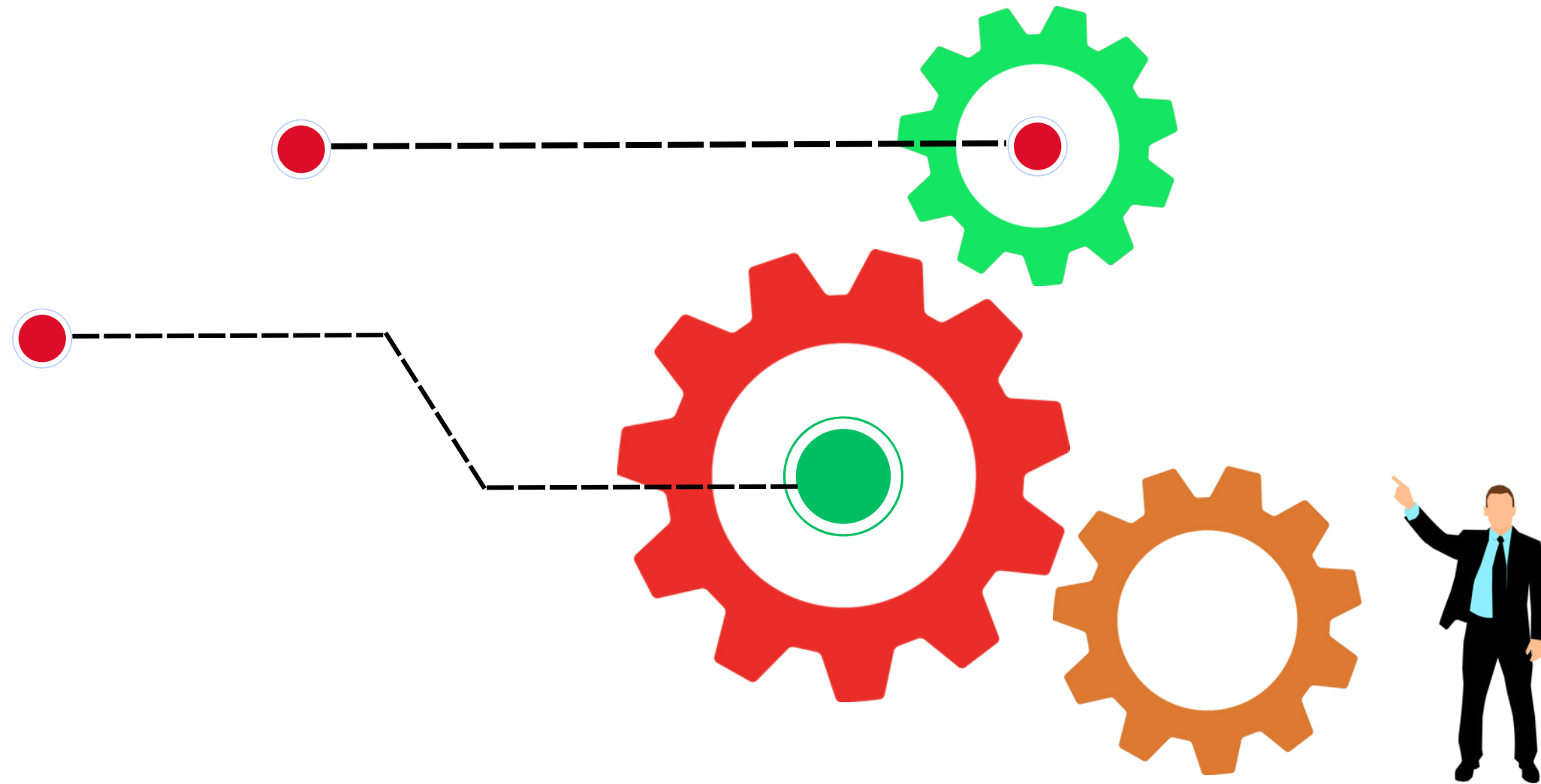
Integration Testing



Integration testing focuses on interactions between components or systems



Integration Testing



- **Component integration testing is often the responsibility of **developers**.**
- **System integration testing is generally the responsibility of **testers**.**

System Testing

- **System testing often produces information that is used by stakeholders to make release decisions**
- **The test environment should ideally correspond to the final target or production environment**
- **System testing should focus on the overall, end-to-end behavior of the system as a whole**
- **Independent testers typically carry out system testing.**



Acceptance Testing



User Acceptance Test

Acceptance testing, like system testing, typically focuses on the behavior and capabilities of a whole system or product

Alpha & Beta Testing



Alpha Testing

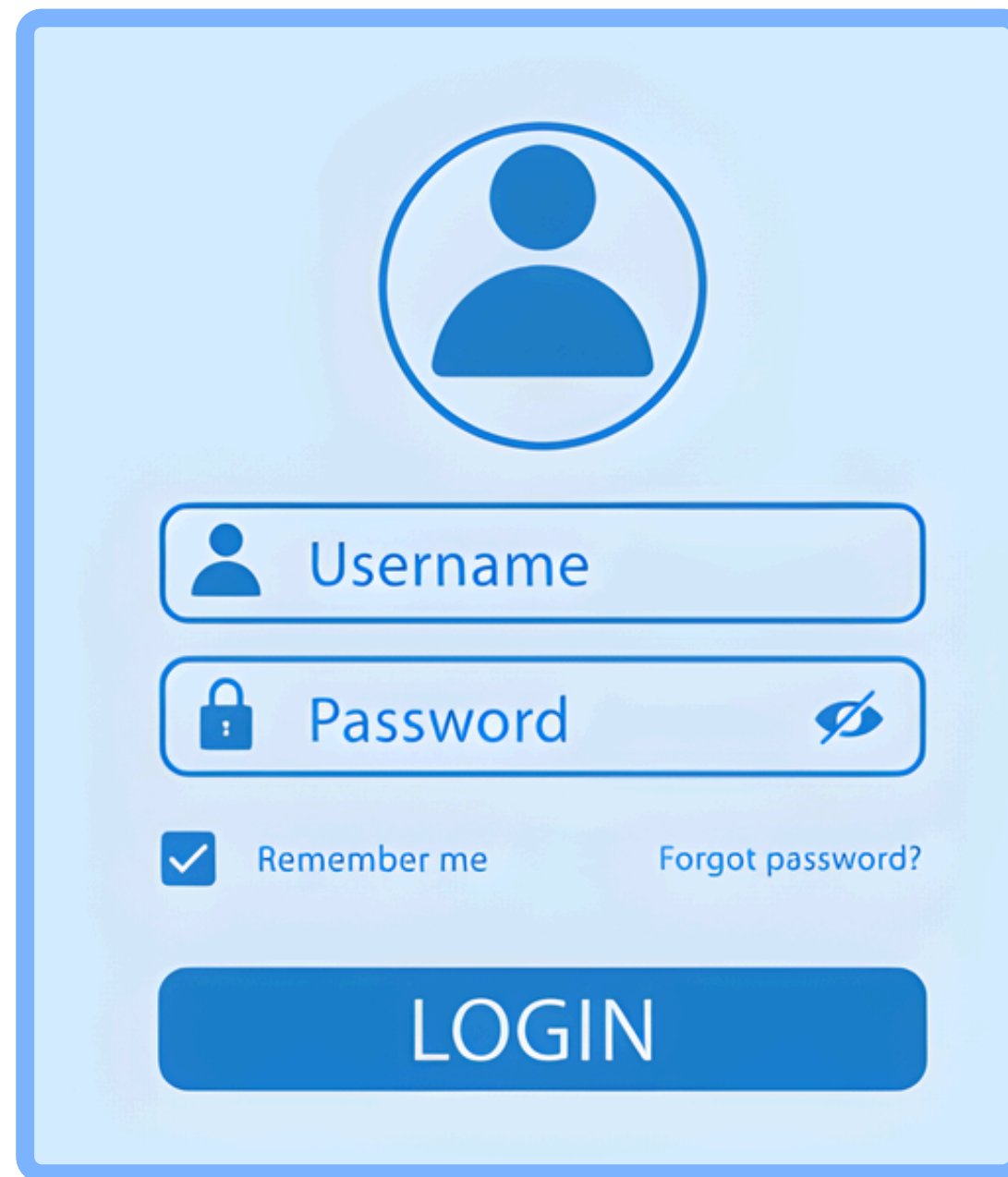


Beta Testing



Testing Types

1-Functional Testing:



A login form interface with a light blue background and rounded corners. At the top center is a circular icon containing a blue silhouette of a person. Below this are two input fields: the first is labeled 'Username' with a small person icon on the left; the second is labeled 'Password' with a lock icon on the left and an eye icon on the right. Below the password field is a checkbox labeled 'Remember me' and a link labeled 'Forgot password?'. At the bottom is a large blue button with the text 'LOGIN' in white capital letters.

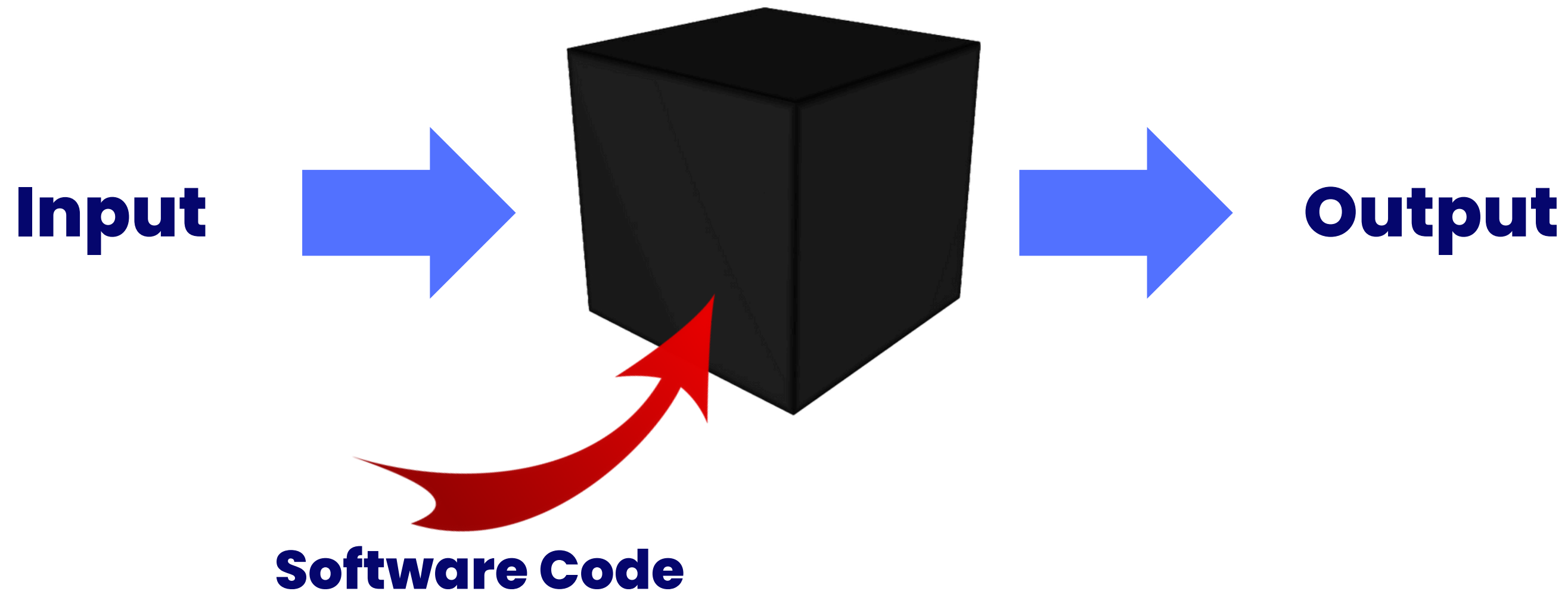
- **Testing what the system does**
- **Usually answered with (Yes/No)**

2-Non-functional Testing



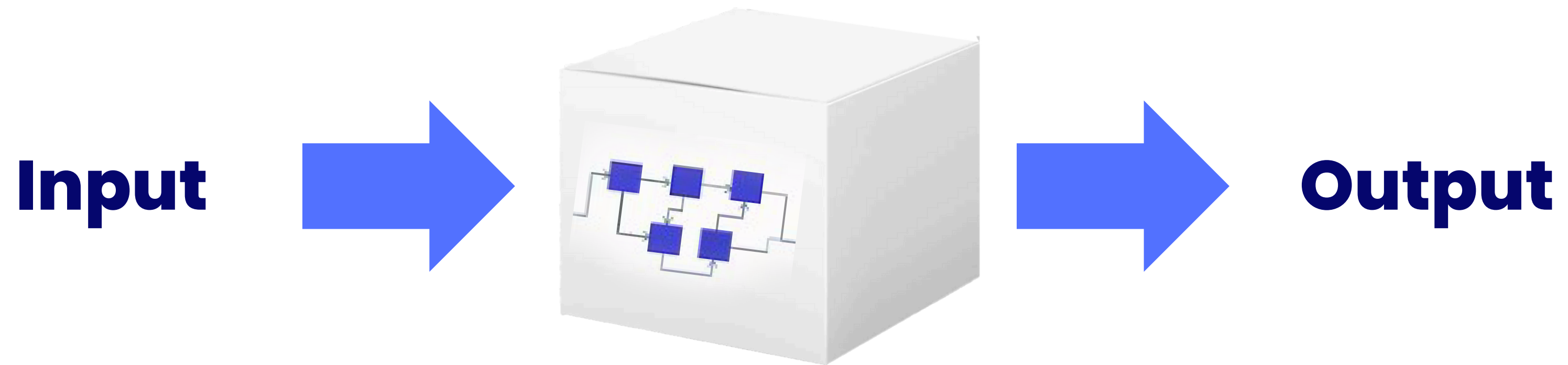
- **Testing how the system performs**
- **Hard to answer with Yes/No**
- **Usually measured as a range**

3-Black-Box Testing:



- **Testing without knowing the internal structure of the system**

4-White-Box Testing:



- **Testing while monitoring the internal structure of the system**

5-Dynamic Testing

The image displays three mobile application screens for dynamic testing, each enclosed in a blue border. The first screen is the 'Sign Up' page, featuring the 'Nezam' logo (a blue 'N' with a red circle) and the text 'Sign Up' in English and Hebrew. It includes a Facebook login button, a text input field, and social media icons for Twitter, YouTube, and LinkedIn. The second screen is the 'Log In!' page, with a blue header and two input fields for email and password. It features a 'Log In' button and social media icons. The third screen is another 'Sign Up!' page, with a blue header and three input fields for name, email, and password. It includes a 'Sign Up' button and social media icons.

- Testing that includes executing the software

6-Static Testing



- Testing that doesn't include executing the software

7-Retesting (Confirmation Testing)



- Testing after debugging to ensure defects are fixed

8-Regression Testing



- Testing unchanged areas to ensure they are not affected by changes

Test Scenario Writing

Test Scenario

- A **TEST SCENARIO** is defined as any functionality that can be tested.
- It is also called Test Condition or Test Possibility.
- As a tester, you should put yourself in the end user's shoes and figure out the real-world scenarios and use cases of the Application Under Test.

How to create a test scenario

1. Carefully study the Requirement Document – Business Requirement Specification (BRS), Software Requirement Specification (SRS), Functional Requirement Specification (FRS) pertaining to the System Under Test (SUT).
2. Isolate every requirement, and identify what possible user actions need to be tested for it. Figure out the technical issues associated with the requirement. Also, remember to analyze and frame possible system abuse scenarios by evaluating the software with a hacker's eyes.
3. Enumerate test scenarios that cover every possible feature of the software. Ensure that these scenarios cover every user flow and business flow involved in the operation of the website or app.
4. After listing the test scenarios, create a Traceability Matrix to ensure that every requirement is mapped to a test scenario.
5. Get the scenarios reviewed by a supervisor, and then push them to be reviewed by other stakeholders involved in the project.



Test Case Writing

What is a test case?



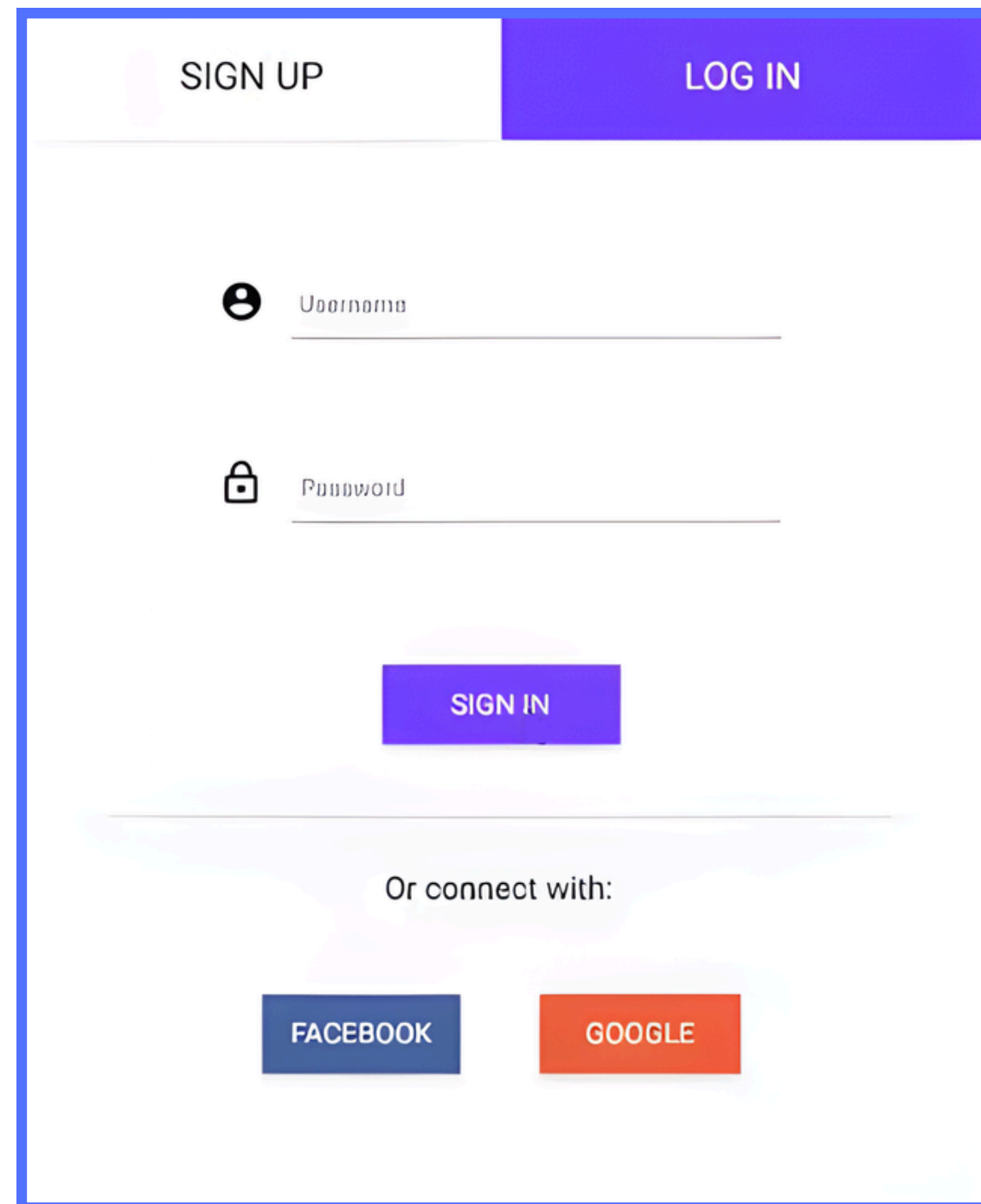
Welcome Admin

☐ Remember Me

[Forgot the password?](#)

- **A set of preconditions, inputs, actions (where applicable), expected results and postconditions, developed based on test conditions.**

1-Test Case Title



SIGN UP LOG IN

Username

Password

SIGN IN

Or connect with:

FACEBOOK GOOGLE



Verify login with a valid username & password

2-Precondition

The image shows a login form with a blue border. At the top, there are two tabs: 'SIGN UP' and 'LOG IN'. The 'LOG IN' tab is highlighted in blue. Below the tabs, there are two input fields: 'Username' with a person icon and 'Password' with a lock icon. A blue 'SIGN IN' button is positioned below the password field. A horizontal line separates the login section from the social login section. Below the line, it says 'Or connect with:'. There are two buttons: 'FACEBOOK' (blue) and 'GOOGLE' (orange).

● **User is already registered using valid credentials**

3-Test Steps

The image shows a login form with a blue border. At the top, there are two tabs: 'SIGN UP' (light blue) and 'LOG IN' (dark blue). Below the tabs, there are two input fields: 'Username' with a person icon and 'Password' with a lock icon. A blue 'SIGN IN' button is positioned below the password field. Below the button, there is a horizontal line and the text 'Or connect with:'. At the bottom, there are two buttons: 'FACEBOOK' (dark blue) and 'GOOGLE' (orange).

- 1. Enter a valid username
- 2. Enter a valid password
- 3. Click on sign in

4-Expected result

The image shows a login form with a blue border. At the top, there are two tabs: 'SIGN UP' and 'LOG IN'. The 'LOG IN' tab is highlighted in blue. Below the tabs, there are two input fields: 'Username' with a person icon and 'Password' with a lock icon. Below these fields is a blue 'SIGN IN' button. Underneath the button is a horizontal line, and below that is the text 'Or connect with:'. At the bottom, there are two buttons: 'FACEBOOK' (blue) and 'GOOGLE' (orange).



**User is logged in successfully
and redirected to (XYZ) page**

5-Test Scenario (Test Suite)

SIGN UP

LOG IN

 Username

 Password

SIGN IN

Or connect with:

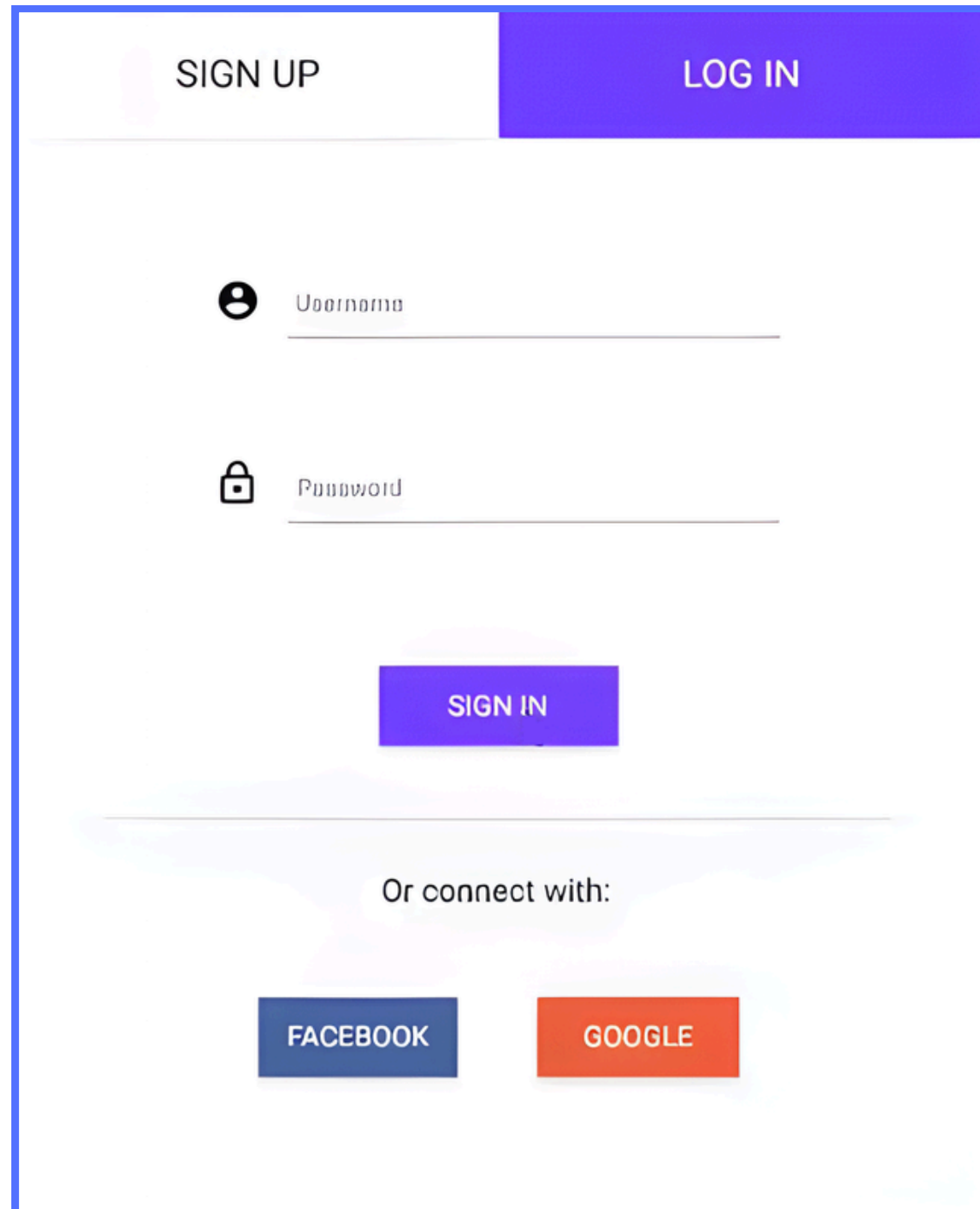
FACEBOOK

GOOGLE



Login

6-Test Environment



SIGN UP LOG IN

Username

Password

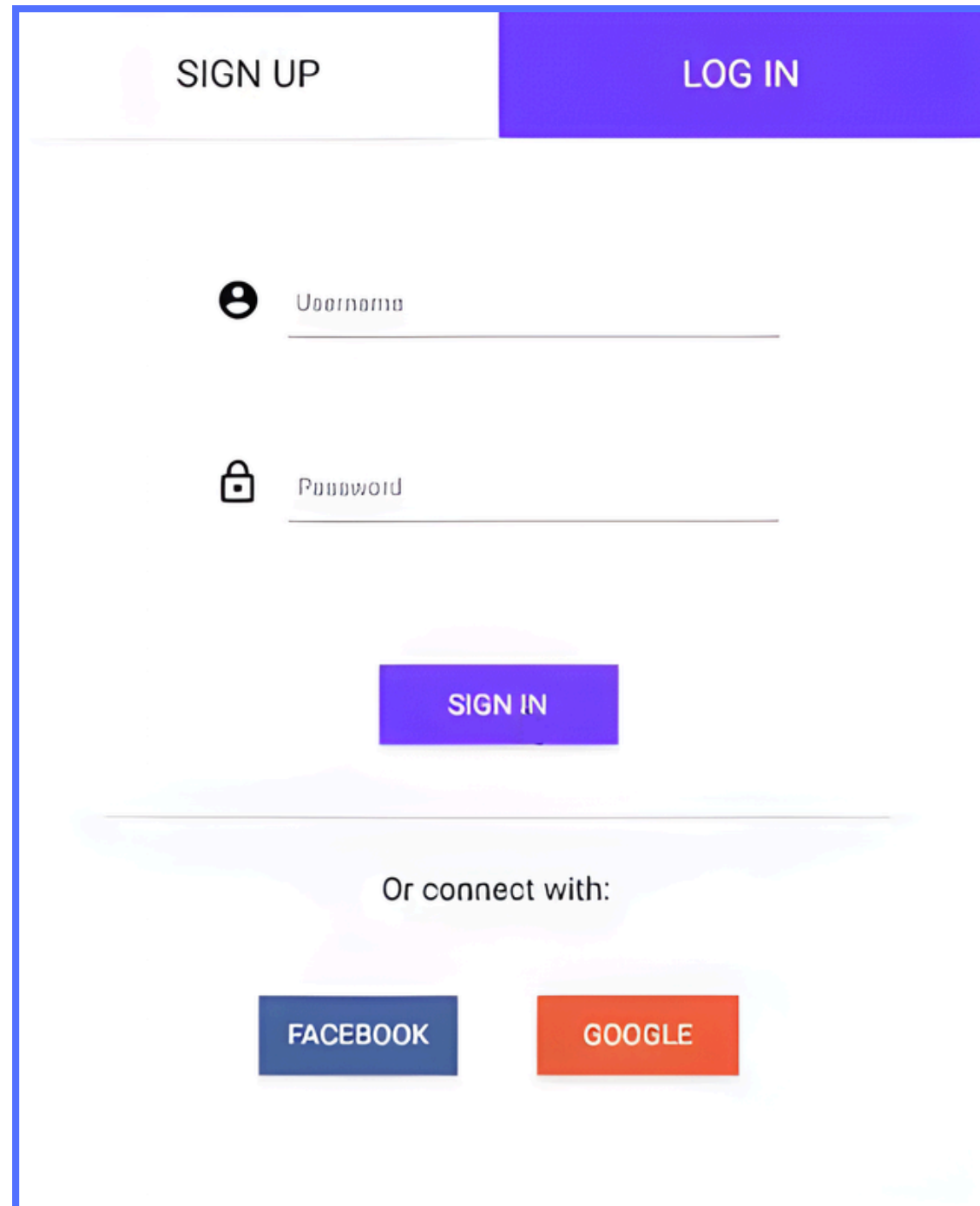
SIGN IN

Or connect with:

FACEBOOK GOOGLE

- **Windows 11 – Chrome – Wi-Fi Samsung**
- **Samsung Note 20 – Android 13 4G Network**
- **iPhone 15 Pro – iOS 16.1 – 5G Network**

7-Actual Result:



The image shows a login interface with a header containing 'SIGN UP' and 'LOG IN' tabs. Below the tabs are two input fields: 'Username' with a person icon and 'Password' with a lock icon. A purple 'SIGN IN' button is positioned below the password field. A horizontal line separates this from a section titled 'Or connect with:' which contains two buttons: 'FACEBOOK' (dark blue) and 'GOOGLE' (orange).

● **Very important note: Never fill the actual result field until you execute the test case**

8-Status

New (Ready to test)	The test case is not executed
Pass	The test case is executed and the actual result is the same as the expected result
Fail	The test case is executed and the actual result is different from the expected result
Blocked/Skipped	The test case can't be executed

Sample Test Case Before Execution

Title	Login with a valid username & password
Precondition	User is already registered using valid credentials
Test Steps	1. Enter a valid username 2. Enter a valid password 3. Click on sign in
Expected Result	-User is logged in successfully -User is redirected to Home page
Test Suite	Login
Test Environment	Samsung Galaxy Note 10 Android 10 – 4G Network
Actual Result	
Status	

Sample Test Case After Execution

Title	Login with a valid username & password
Precondition	User is already registered using valid credentials
Test Steps	1. Enter a valid username 2. Enter a valid password 3. Click on sign in
Expected Result	-User is logged in successfully -User is redirected to Home page
Test Suite	Login
Test Environment	Samsung Galaxy Note 10 Android 10 – 4G Network
Actual Result	Same as expected
Status	Pass

Let's now create test cases for different functionalities

- **Valid/Invalid Sign Up Cases**
- **Valid/Invalid Login Cases**
- **Valid/Invalid Search Cases**

Test Case Writing Tools



Qase



TestLodge



XRAY

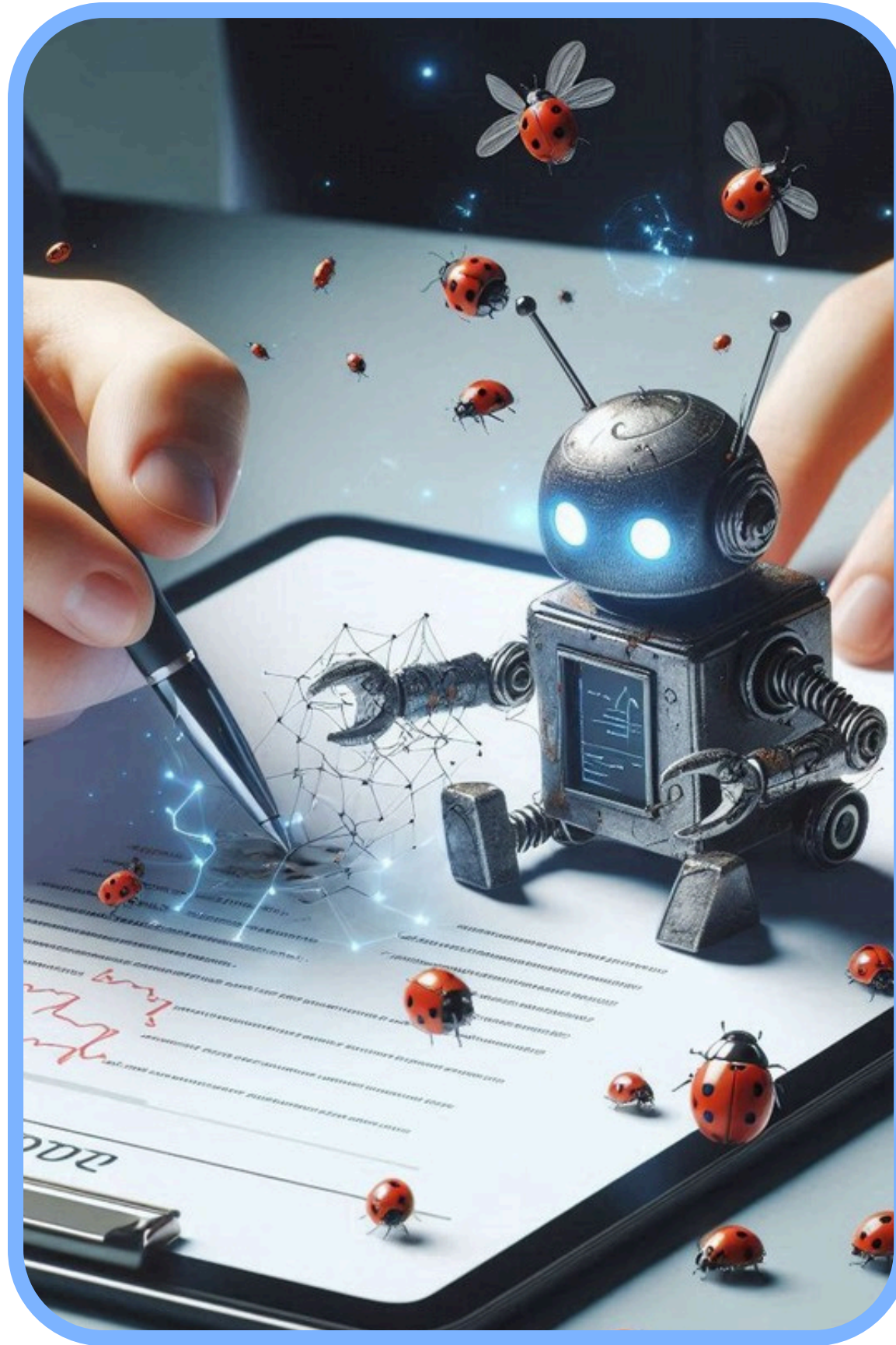
TestRail



Test Case Writing using



Test Execution & Defect Reporting



How to write a Defect Report

What is a defect report?

- Documentation of the occurrence, nature, and status of a defect.

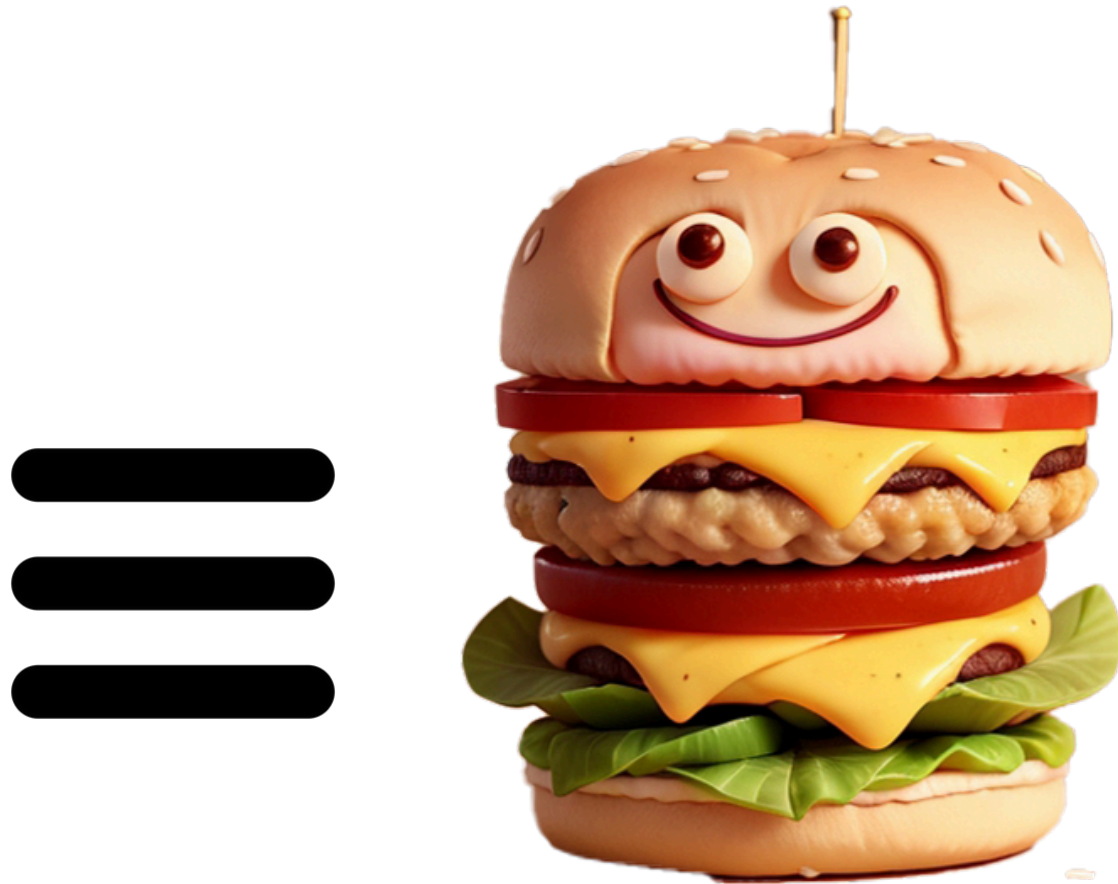
What is a defect ?

- An imperfection or deficiency in a work product where it does not meet its requirements or specifications.
[Synonyms: bug, fault]

1-Defect Report Title:

- Section -> Description
- Register -> No error message appears when user leaves password field empty

2-Steps to reproduce:



- Must be very specific
- Ex:
 1. Open www.nezamacademy.com
 2. Click on the hamburger icon in the upper left corner
 3. Scroll down to the bottom of the screen
 4. Click on Settings
 5. Click on data usage
 6. Change data usage to "minimum limit"

3-Expected Result:

- The same expected result like in the test case

4-Actual Result:

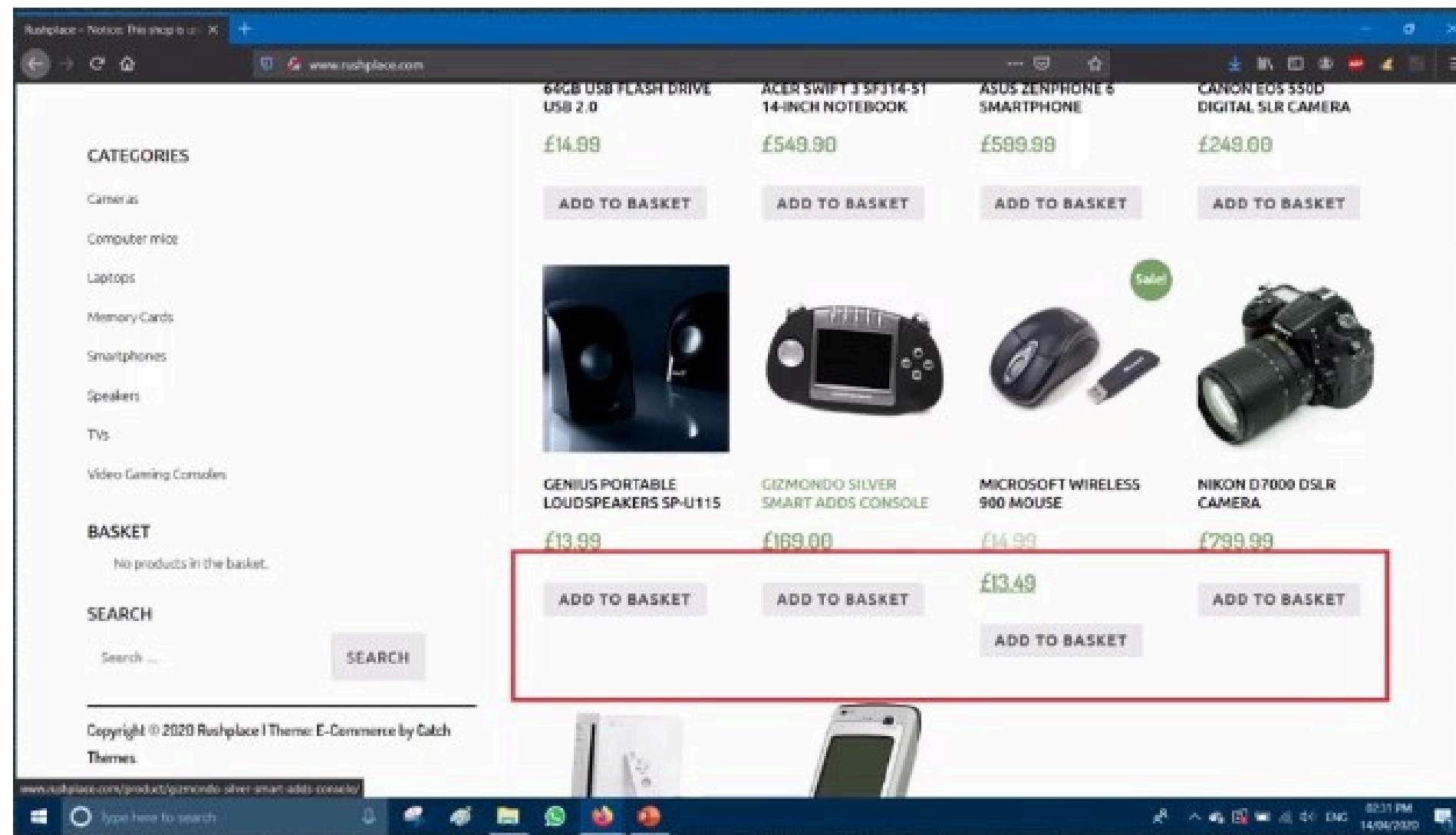
- What really happened when the steps are executed

5-Test Environment

- Specify the environment(s) in which the defect occurred :
 - Windows 11 – Chrome – Ethernet – Development environment
 - Macbook Pro M1 – Wi-Fi
 - iPhone 11 iOS 13.3.1 5G Network – Production environment

6-Screenshot or video:

- The screenshot must show the whole screen
- A red rectangle should be around the defect area
- The video should show the clicks on the keyboard



Click



Tap



7-Defect priority:

- **Critical:** Login isn't working-Application crashes in Home page - Wrong cart value
- **High:** Login page responds slowly - User is not able to add profile image
- **Medium:** Some pages have poor performance- Portrait mode isn't working correctly
- **Low:** Spelling mistakes - Image misalignment



Types of Defects

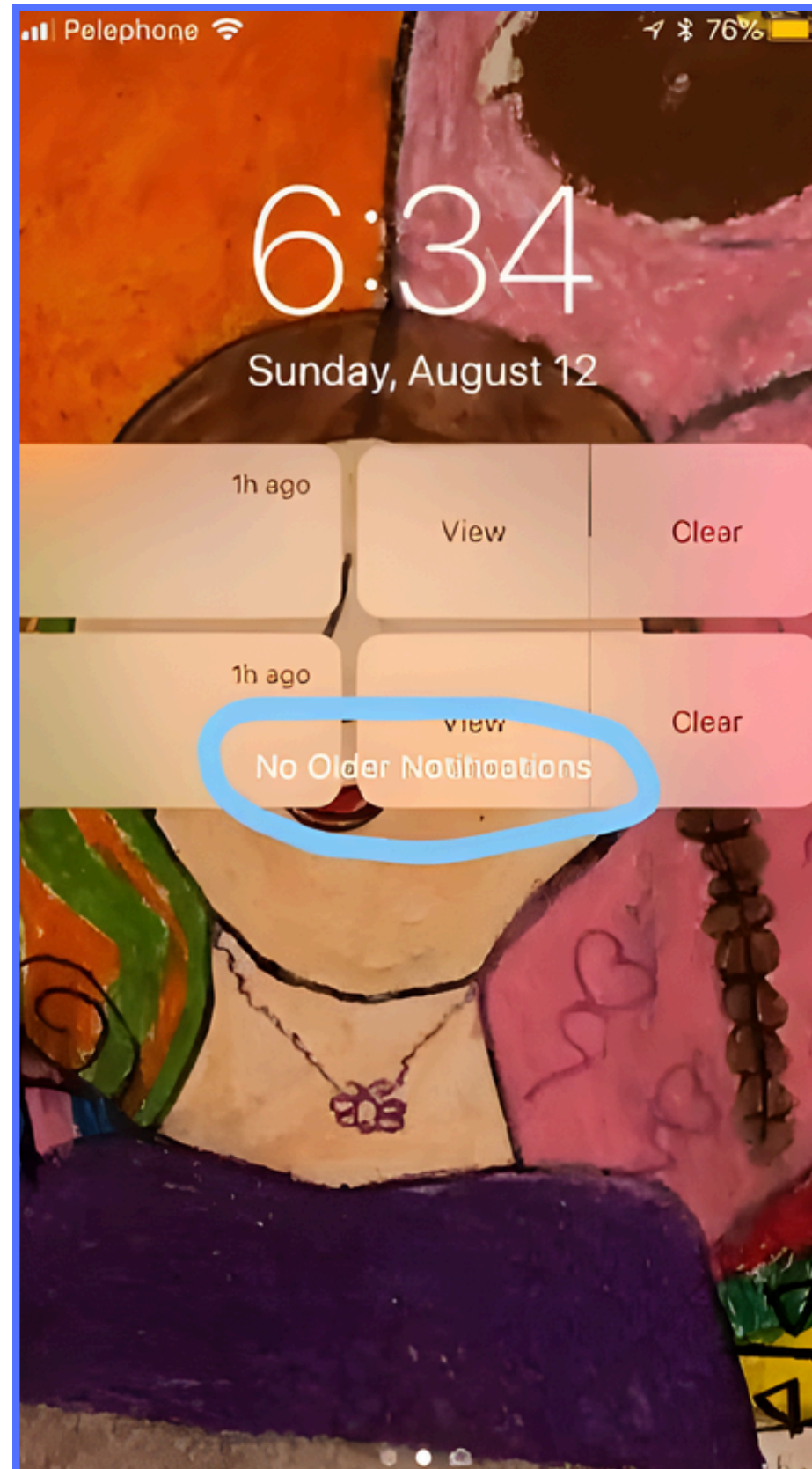
1-Functional:



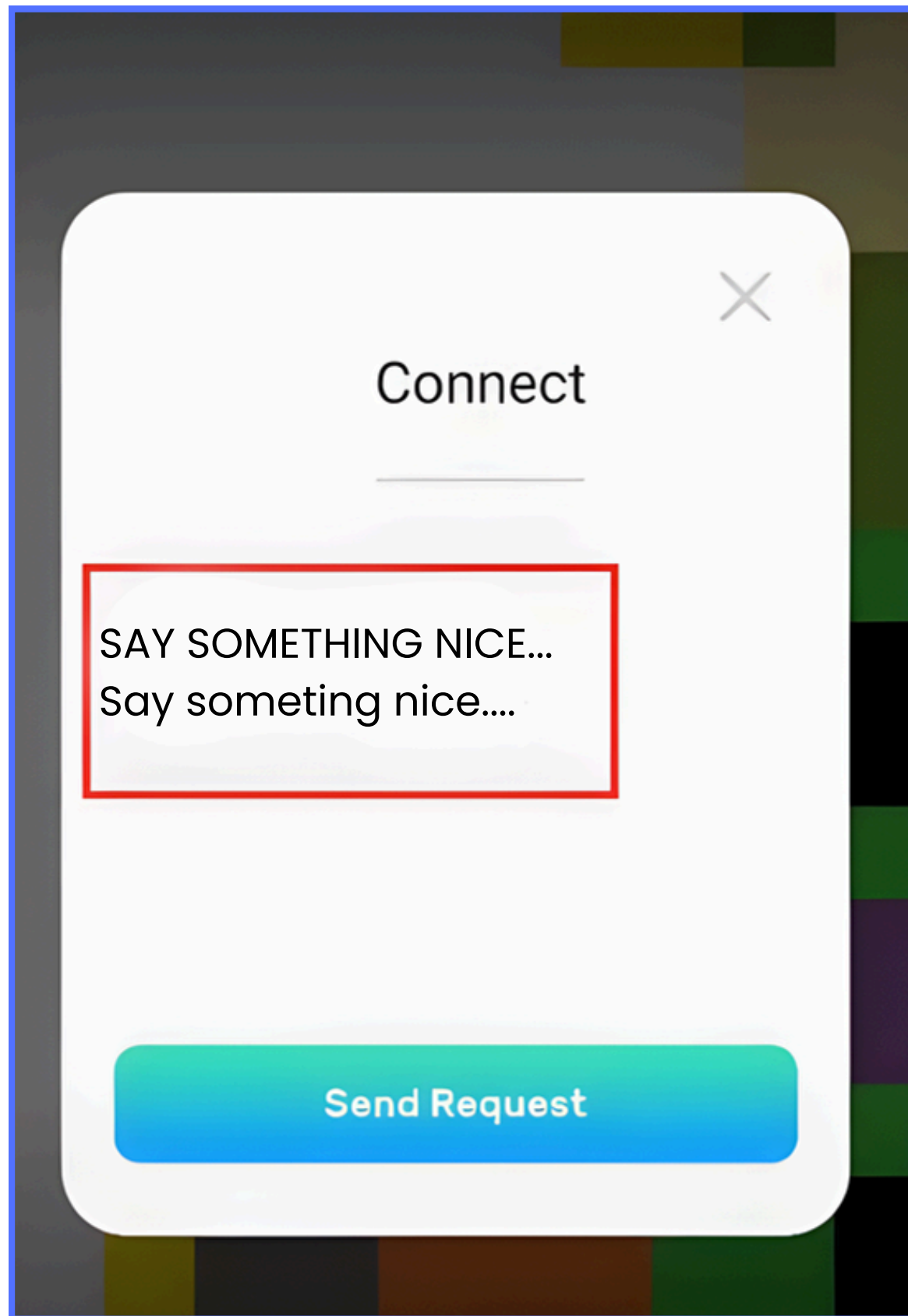
Forgot password functionality isn't working

The screenshot shows the WEGLOT login interface. At the top, the text "WEGLOT" is displayed in a bold, dark font. Below it, the instruction "LOG IN TO YOUR ACCOUNT" is centered. The login form consists of two input fields: "Email Address" and "Password", each with a small icon to its right. Below these fields is a green "LOG IN" button. Underneath the button is a link labeled "Forgot your password?", which is highlighted with a red rounded rectangle. A red arrow points from the right side of the image towards this link. At the bottom of the form, there is a link that says "No account yet ? Sign up".

2-Visual (UI):



3-Content:

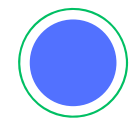


Connect

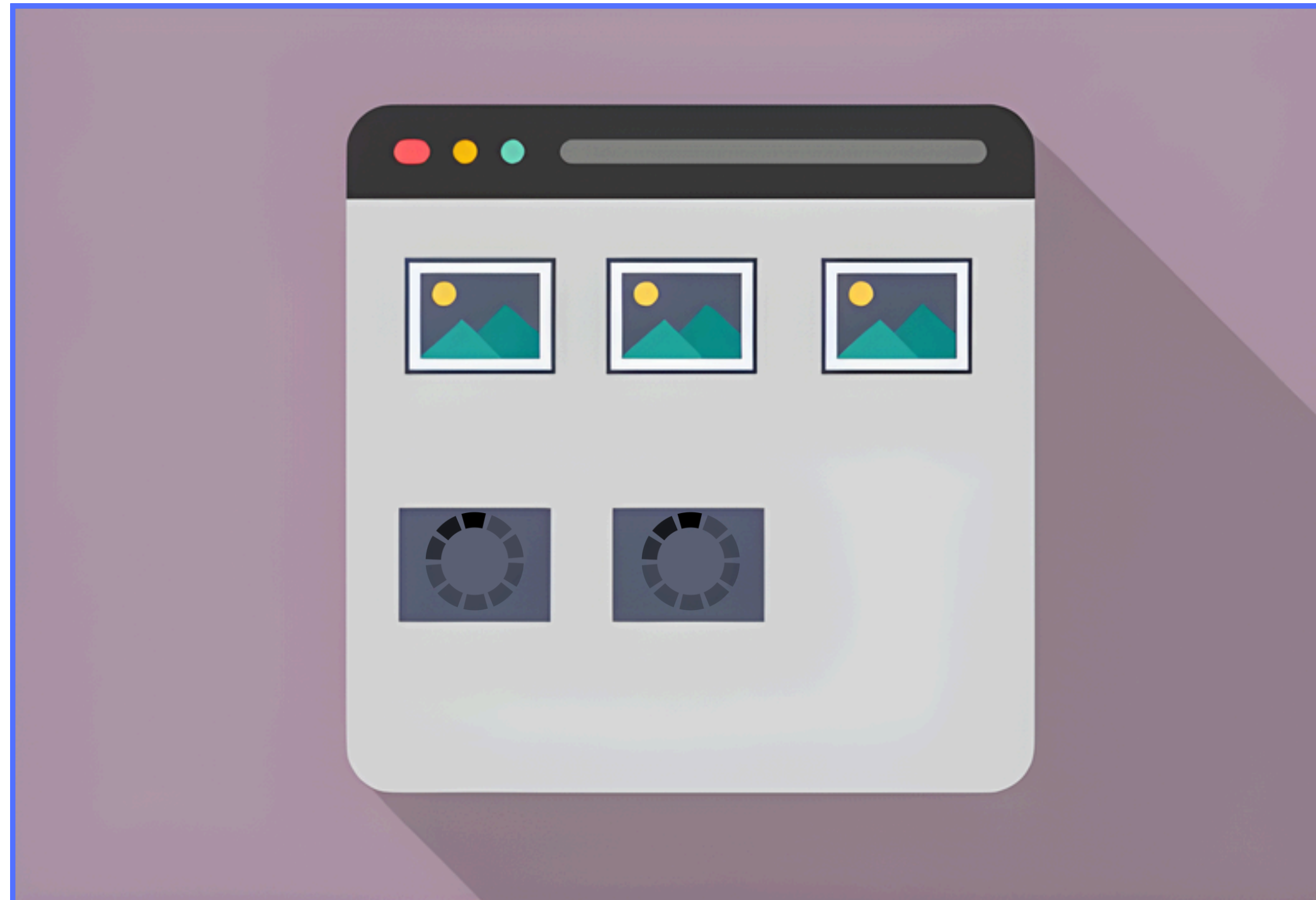
SAY SOMETHING NICE...
Say someting nice....

Send Request

4-Performance:



Videos take too much time to play



5-Suggestion:

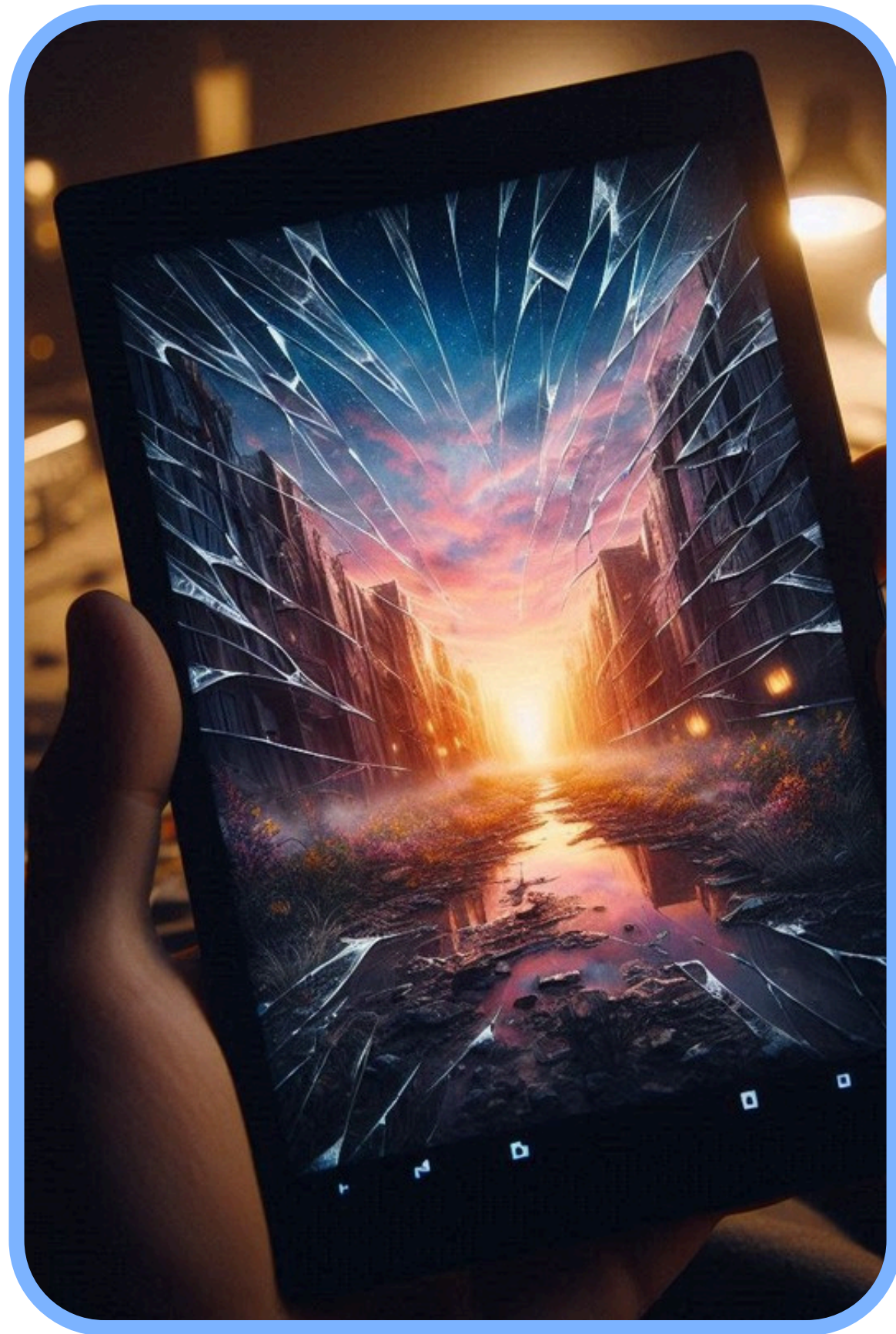


The font of the placeholder should be bigger



Example Defect Report

Title	Login → Forgot password button isn't working
Steps to reproduce	1. Click on Login 2. Click on Forgot password
Expected Result	The button can be clicked and user should be redirected to a page to enter his email
Actual result	Clicking on the button doesn't have any impact
Test Environment	Samsung Galaxy Note 10 – Android 10 – 4G Network
Priority	High
Type	Functional
Screenshot	



Taking Screenshots for the Defect Report



Video Recording for the Defect Report

OverviewPlatform featuresLog streaming

ReconnectCopy logsPauseClear

Application logsWeb Server logs

Connecting...
2018-10-10T06:15:54 Welcome, you are now connected to log-streaming service.
2018-10-10T06:16:54 No new trace in the past 1 min(s).
2018-10-10T06:17:54 No new trace in the past 2 min(s).
2018-10-10T06:18:54 No new trace in the past 3 min(s).
2018-10-10T06:19:54 No new trace in the past 4 min(s).
2018-10-10T06:20:54 No new trace in the past 5 min(s).
2018-10-10T06:21:54 No new trace in the past 6 min(s).
2018-10-10T06:22:54 No new trace in the past 7 min(s).
2018-10-10T06:23:54 No new trace in the past 8 min(s).
2018-10-10T06:24:55 No new trace in the past 9 min(s).
2018-10-10T06:25:55 No new trace in the past 10 min(s).
2018-10-10T06:26:55 No new trace in the past 11 min(s).
2018-10-10T06:27:55 No new trace in the past 12 min(s).
2018-10-10T06:28:55 No new trace in the past 13 min(s).
2018-10-10T06:29:55 No new trace in the past 14 min(s).
2018-10-10T06:30:55 No new trace in the past 15 min(s).
2018-10-10T06:31:55 No new trace in the past 16 min(s).
2018-10-10T06:32:55 No new trace in the past 17 min(s).
2018-10-10T06:33:55 No new trace in the past 18 min(s).

Capturing Web Logs



Accessibility Defects



Will you report those defects?



Application includes a functionality that contradicts with requirements



Application misses a functionality that is included in requirements



Application includes a functionality that is not mentioned in requirements



Application misses an essential requirement that is not included in requirements



Application Look & Feel

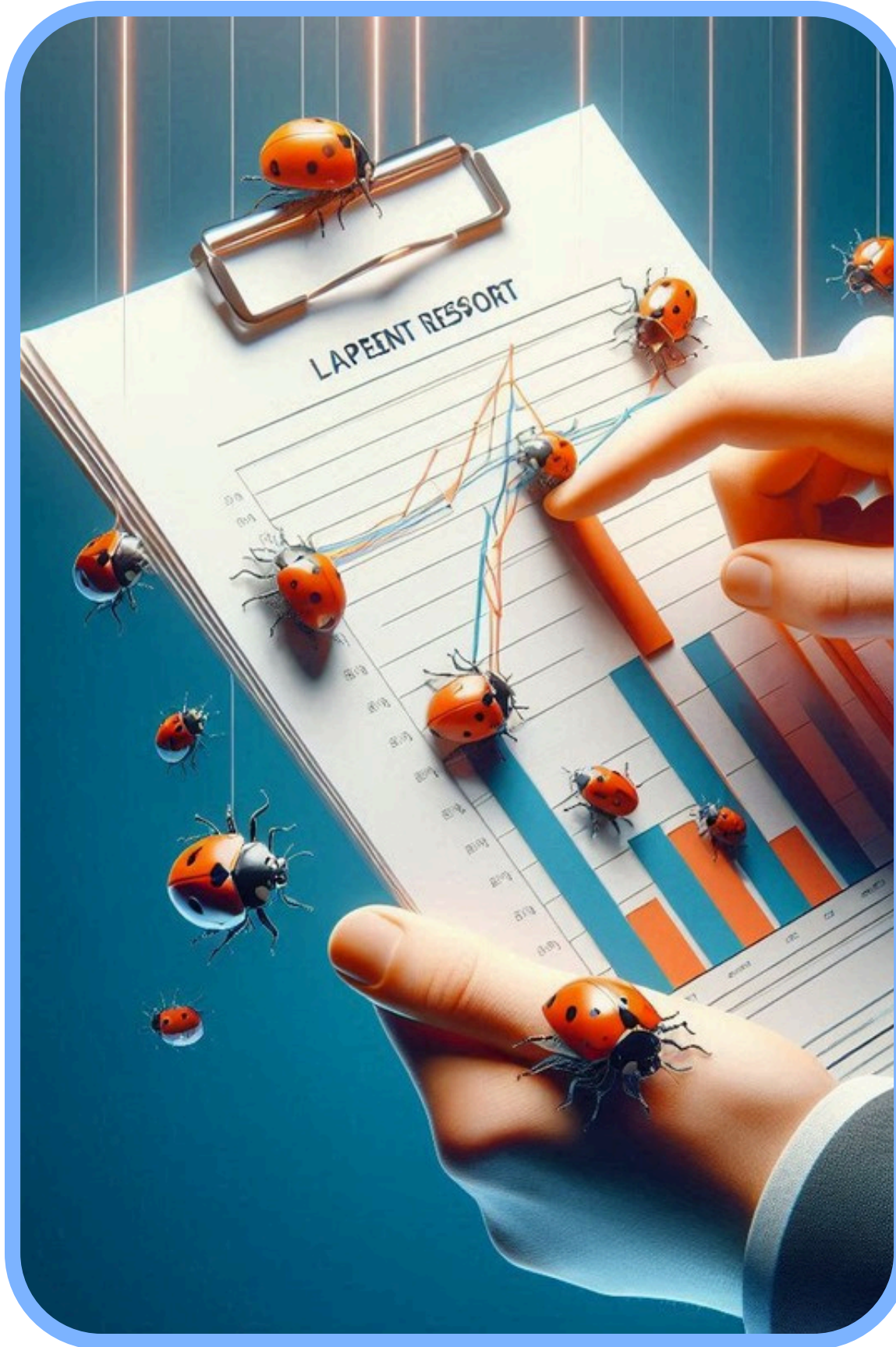
WHAT ABOUT CONTRADICTION BETWEEN REQUIREMENTS AND DESIGN ?



Testing Reports: Test Plan, Progress Report, & Completion Report



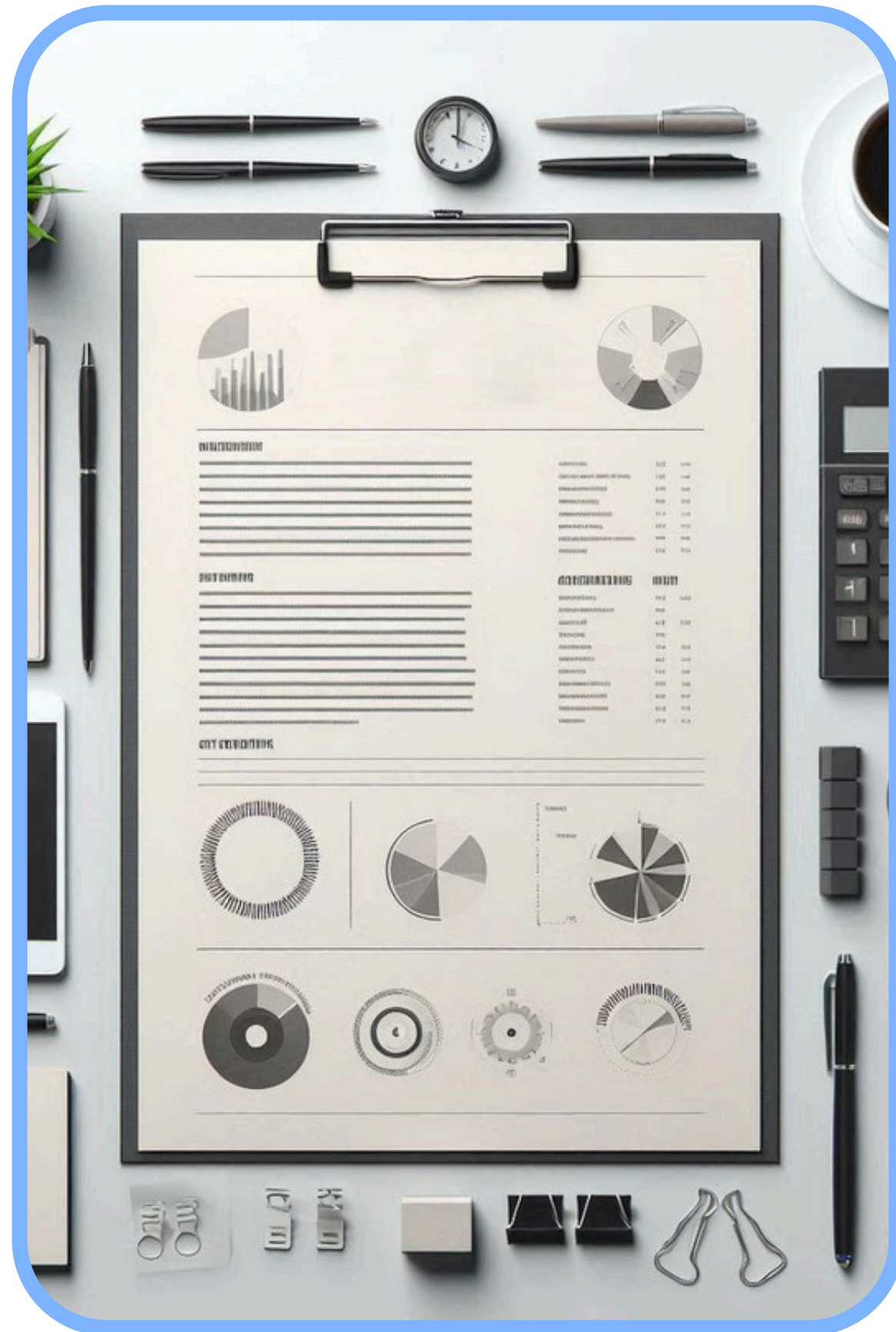
**Who is your
audience?**



Test Reports

Types of Test Reports

- **Test Progress report:** A type of test report produced at regular intervals about the progress of test activities against a baseline, risks, and alternatives requiring a decision. Synonyms: test status report
- **Test completion report:** A type of test report produced at completion milestones that provides an evaluation of the corresponding test items against exit criteria.



Sample Testing Report & How to write it